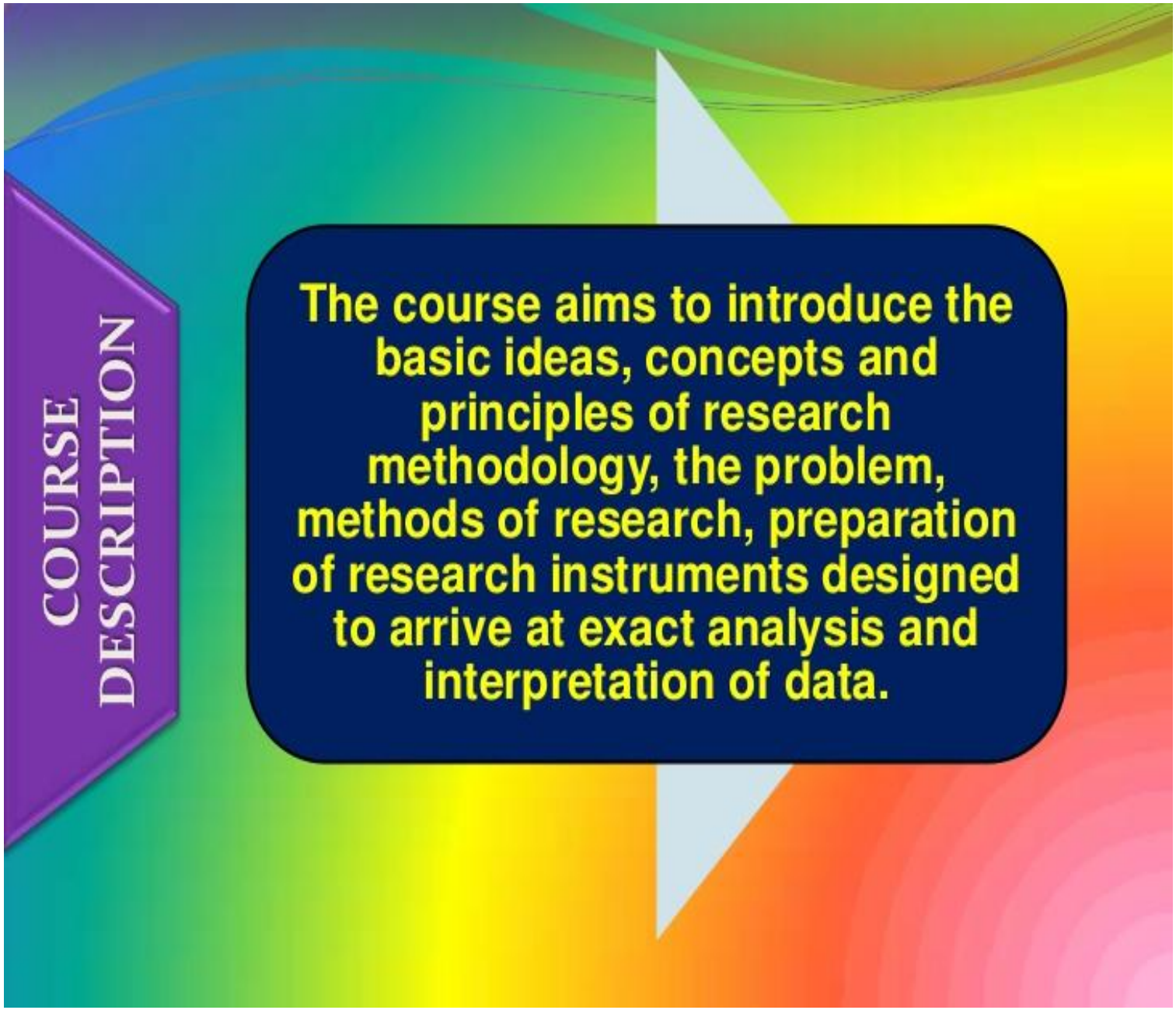
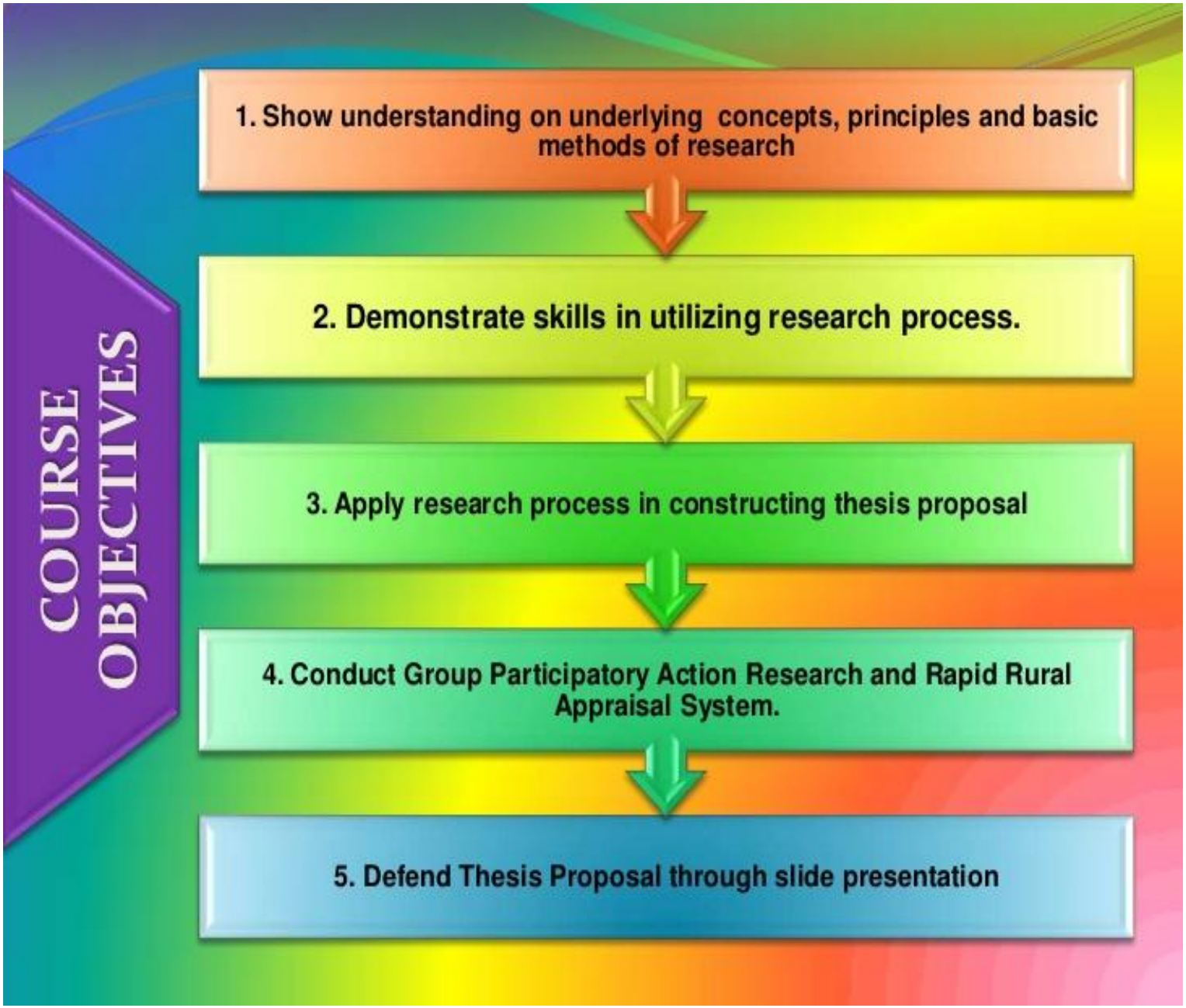


METHODS OF RESEARCH




COURSE DESCRIPTION
The course aims to introduce the basic ideas, concepts and principles of research methodology, the problem, methods of research, preparation of research instruments designed to arrive at exact analysis and interpretation of data.


COURSE OBJECTIVES
1. Show understanding on underlying concepts, principles and basic methods of research
2. Demonstrate skills in utilizing research process.
3. Apply research process in constructing thesis proposal
4. Conduct Group Participatory Action Research and Rapid Rural Appraisal System.
5. Defend Thesis Proposal through slide presentation


COURSE CONTENT
Introduction of Research
Research Problems & Objectives
Related Literature and Studies
Research Methods & Design
Sampling Design
Collection of Data
Data Processing
Statistical Treatment
Data Analysis/ Presentation & Interpretation
Summary/ Conclusion & Recommendations
Bibliography
Thesis Proposal Writing

INTRODUCTION TO RESEARCH

1.1. Definition, Purpose, characteristics,
differences, kinds and classification of Research
Qualities of Good Research
Qualities of Good Researcher
Hindrances of scientific Inquiry,
Major needs and problems/demands of Research
Values of Research to Man
Types of Research
Classification of Research,
The variable
Components of Research Process,

1.2. Qualities of Good Research
Instrument
Validity, Types of validity, reliability,
methods in testing the reliability of research instrument,
usability
determining factors of usability of research

THE RESEARCH PROBLEM & OBJECTIVES

2.1

- The Research Problems, characteristics of a Research Problem, Sources of Research Problem
- The Research Objectives,
- Statement of the Problem/Objectives,
- Hypothesis, Theoretical and Conceptual Framework, Assumptions,
- Significance of the Study, Scope and Delimitation of the Study

RELATED LITERATURE & STUDIES

- 3.1.Importance of Related Literature and Studies
- 3.2, Purpose and Functions of related studies,
- 3.3 Characteristics,
- 3.4. Sources and Where to locate the sources,
- 3.5 Related Legal Bases, Related Literature, Related Studies(Local, Foreign)
- 3.6. Definition of Terms: Conceptual and Operational

RESEARCH METHODS/ RESEARCH DESIGNS

- 4.1.Historical Design:

  Uses of historical research, major steps of historical designs, sources of historical research

- 4.2.Descriptive Design: Meaning, characteristic, value, importance and advantages, techniques, Types of Descriptive Research : Descriptive- Survey, Descriptive-normative survey, Descriptive-status, Descriptive analysis, Descriptive classification, descriptive-evaluative, descriptive-comparative, correlational survey, longitudinal survey

- 4.3. Experimental Design: Meaning, concepts of causation, distinguishing features, experimental designs, experimental plan, Types of Experimental Designs: Single-group design, Two-group design, two-pair group design, parallel group design, counterbalanced or latin square design, complete randomized design, randomized complete block design, correlational Design, Pre-test and Post test design, case study design, Case work, Content Analysis

SAMPLING DESIGNS

- 5.1. Definition, Purpose, Terminologies, Principles, advantages and Disadvantages of Sampling, Planning a Sample Survey,
- 5.2. Determination of Sample Size
- 5.3. Sampling Design:
  (A) Scientific Sampling:
  Restricted random sampling, unrestricted random sampling, stratified random sampling, systematic sampling, multistage sampling, cluster sampling
  (B) Nonscientific sampling:
  Purposive Sampling, Incidental sampling, Quota sampling

COLLECTION OF DATA

6.1. Classification of Data According to Source, Selecting method of collecting Data, Research Instruments or Tools

6.2. The Questionnaire: Definition, Advantages and Disadvantages, Construction of Questionnaire, Types of Questions, Guidelines in Formulation, Problems in responses, evidence of misleading questions, the cover letter, evaluating the questionnaire, sample questionnaires

6.3. The Interview: Purpose and uses of interview, advantages and disadvantages, types of classes of interviews, interview as instruments, steps/ pointers, what to avoid in interview

DATA PROCESSING

- 7.1. Data Processing: Definition
- 7.2. Data Matrix: Types of matrices (Univariate matrix, bivariate, multivariate)
- 7.3. Dummy tables

STATISTICAL TREATMENT

8.1. Univariate Statistical Treatment, Bivariate Statistical Treatment, Multivariate Statistical Treatment

8.2. Frequency Distribution and Graphical Representation

8.3. Descriptive Statistics: Measures of Central Tendency, Measures of Variability, Normal Distribution of Standard, Measures of Relative Variability, Measures of Kurtosis, Measures of Skewness

8.4. Inferential Statistics: Correlation,

- Chi-square, z test means, t-test, f-test, Kruskal-Wallis One-way
- Analysis of variance by ranks,
- Friedman Two-way analysis of variance by ranks

DATA ANALYSIS, PRESENTATION AND INTERPRETATION OF DATA

9.1. Data Analysis: Types of Data

- Analysis (Univariate Analysis, Bivariate Analysis, Multivariate Analysis, Normative Analysis, Status Analysis, Descriptive Analysis, Classification Analysis, Evaluative Analysis, Comparative Analysis)

9.2. Presentation of Data:

- Textual presentation, tabular presentation,
- Graphical Presentation, Purpose of Graphing, Advantages and Limitations of Graphs,
- Types of graphs, construction of Individual Graphs

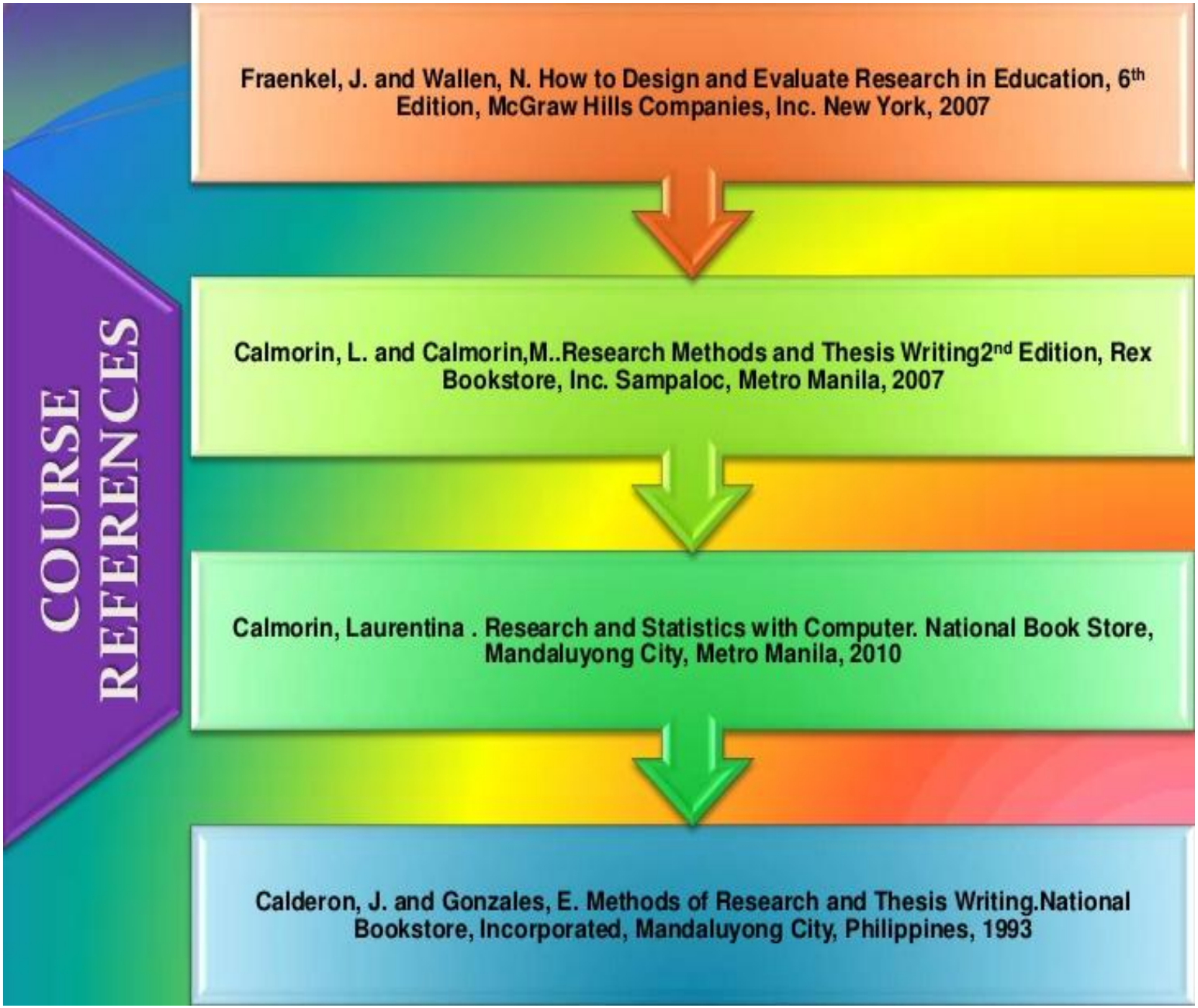
9.3. Group Derive Generalization

SUMMARY, CONCLUSIONS AND RECOMMENDATIONS

- 10.1. Summary of Findings
- 10.2. Conclusion and Some dangers to avoid in drawing up conclusions
- 10.3. Recommendations

THESIS PROPOSAL WRITING

- 11.1. Writing Requirements
- 11.2. Content and Structure of Thesis Proposal: The Problem: Rationale and Background, Review of Related Literature, Methodology,Materials and methods, results and discussion, summary, conclusion and recommendations, Schedule of Activities, Bibliography, Curriculum Vitae
- 11.3. Form and Styles of a Research Paper, Thesis and Dissertations


COURSE REFERENCES
Fraenkel, J. and Wallen, N. How to Design and Evaluate Research in Education, 6th Edition, McGraw Hills Companies, Inc. New York, 2007
Calmorin, L. and Calmorin,M..Research Methods and Thesis Writing2nd Edition, Rex Bookstore, Inc. Sampaloc, Metro Manila, 2007
Calmorin, Laurentina . Research and Statistics with Computer. National Book Store, Mandaluyong City, Metro Manila, 2010
Calderon, J. and Gonzales, E. Methods of Research and Thesis Writing.National Bookstore, Incorporated, Mandaluyong City, Philippines, 1993

DEFINITION OF RESEARCH

Research is a careful, critical, disciplined inquiry, varying in techniques and methods according to the nature and conditions of the problem, directed towards clarification or resolution of the problem (Good,1972)

Research is a process of gathering data or information to *solve a particular or specific problem in a scientific manner* (Manuel and Medel,1976)

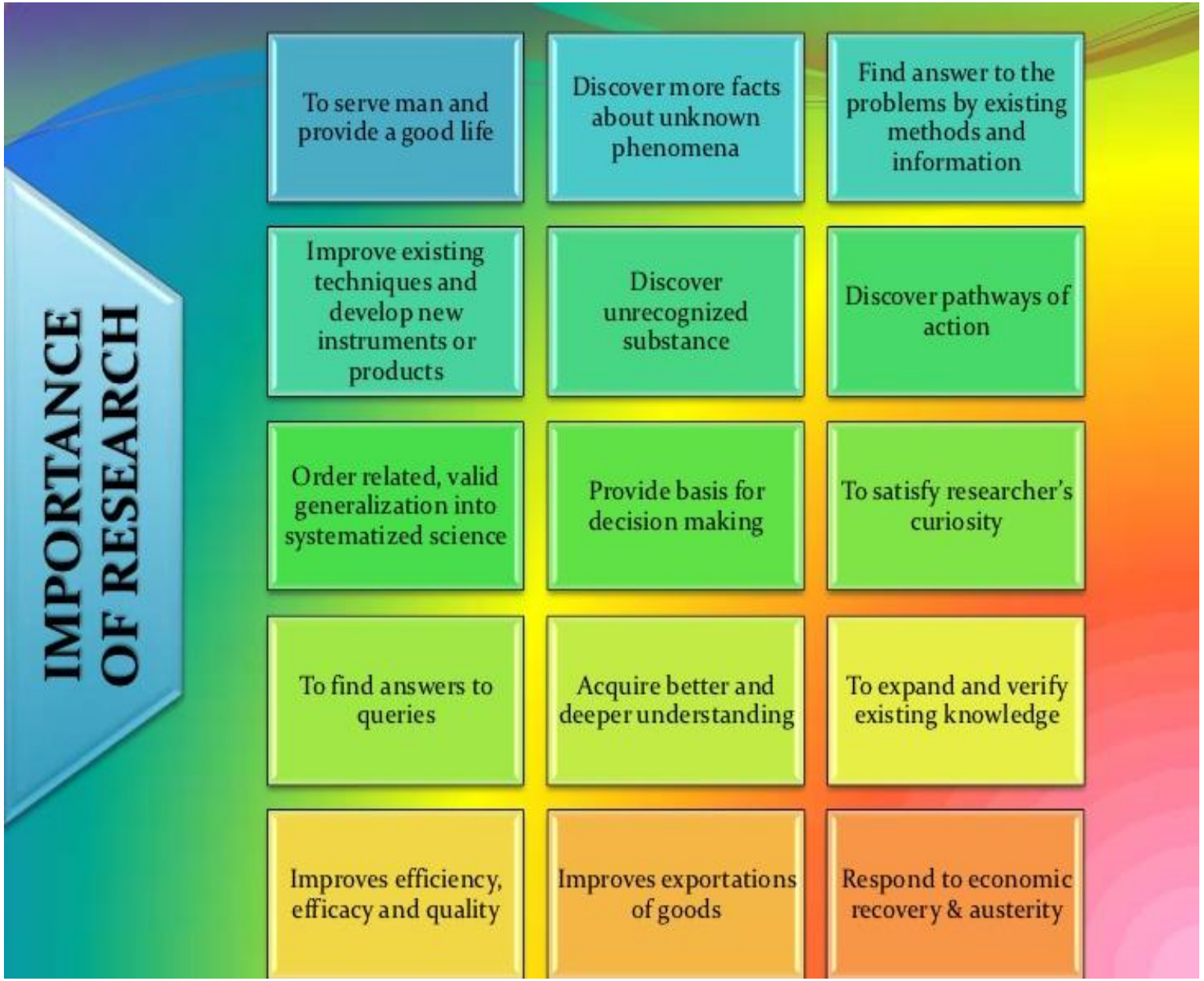
Research is an attempt to *gain solution to the problems,* a collection of data in a rigorously controlled situation for the purpose of prediction or explanation (Treece and Treece, 1973)

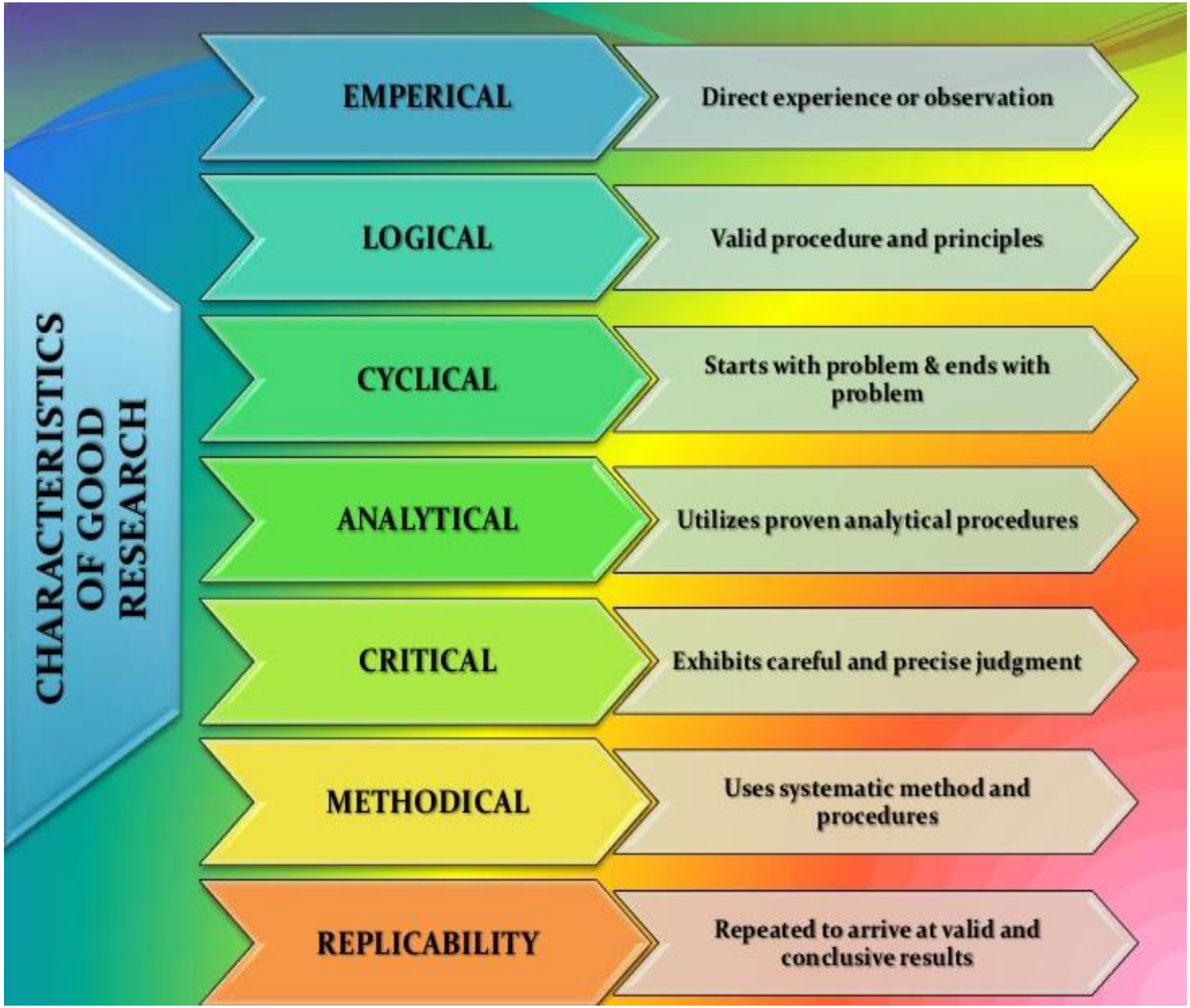
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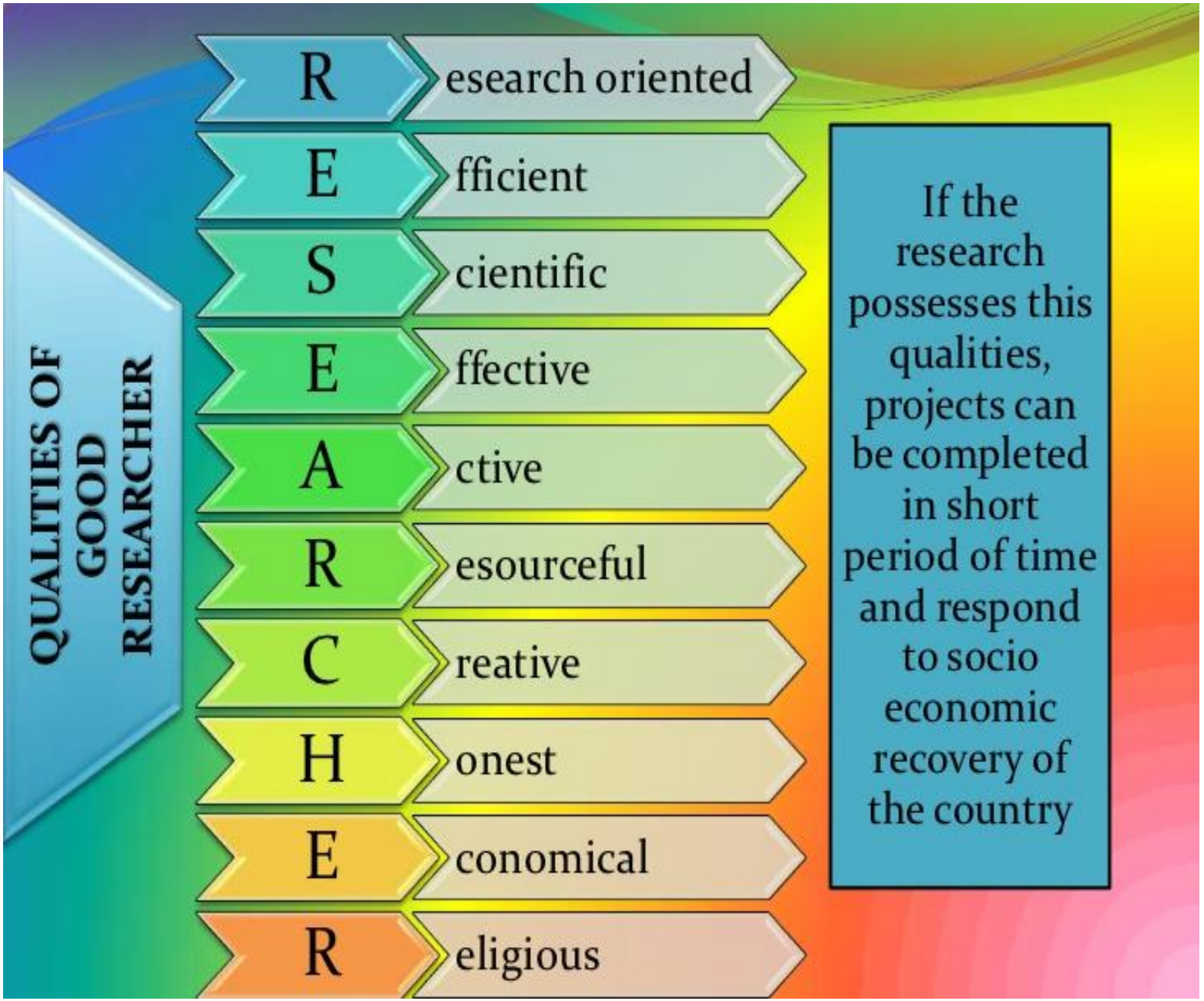
Research is a systematic study or *investigation of something for the purpose of answering questions posed by the* researcher (Sanchez, 1980)

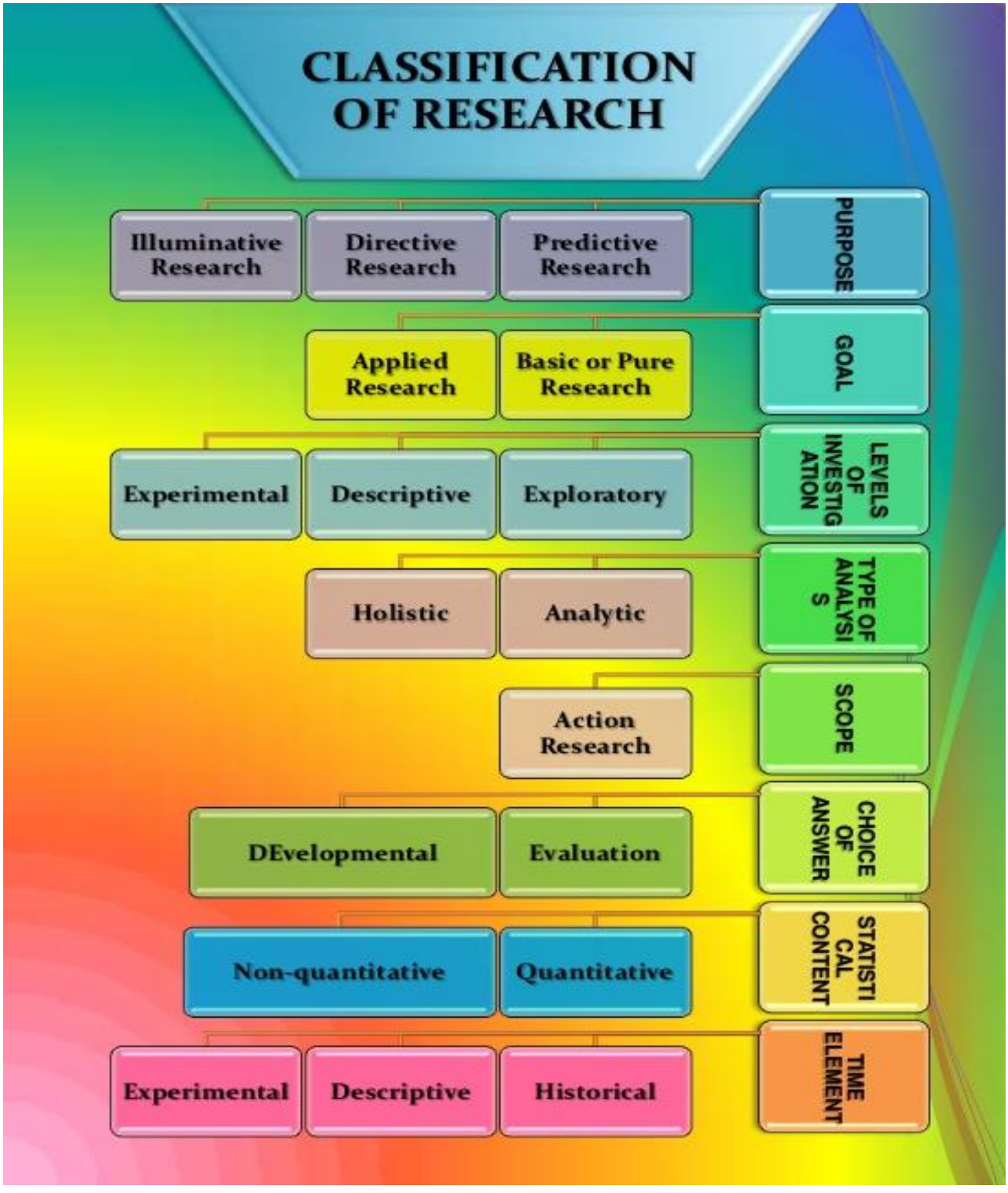
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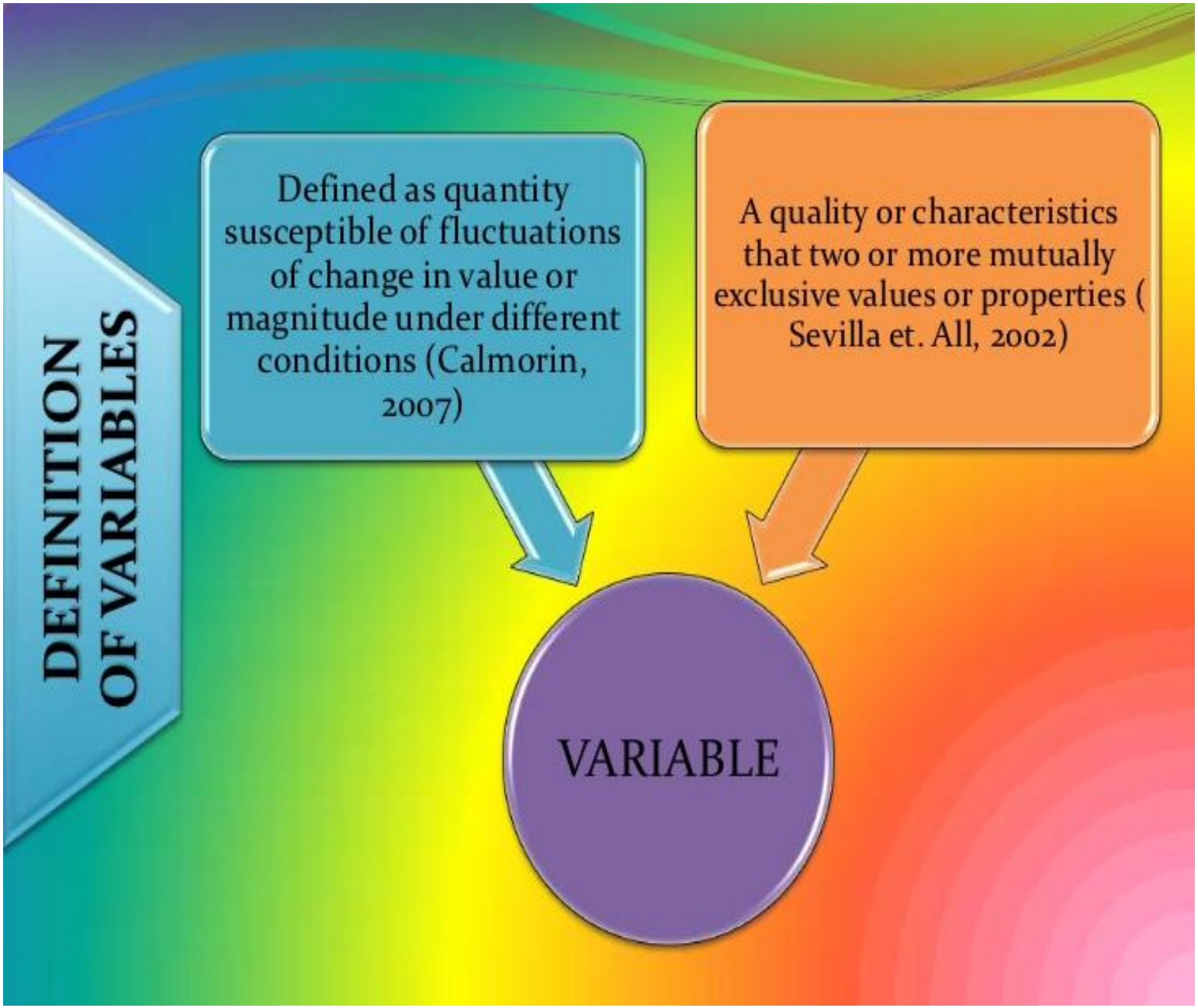
Research is an ORGANIZED and SYSTEMATIC way of FINDING ANSWERS to QUESTIONS

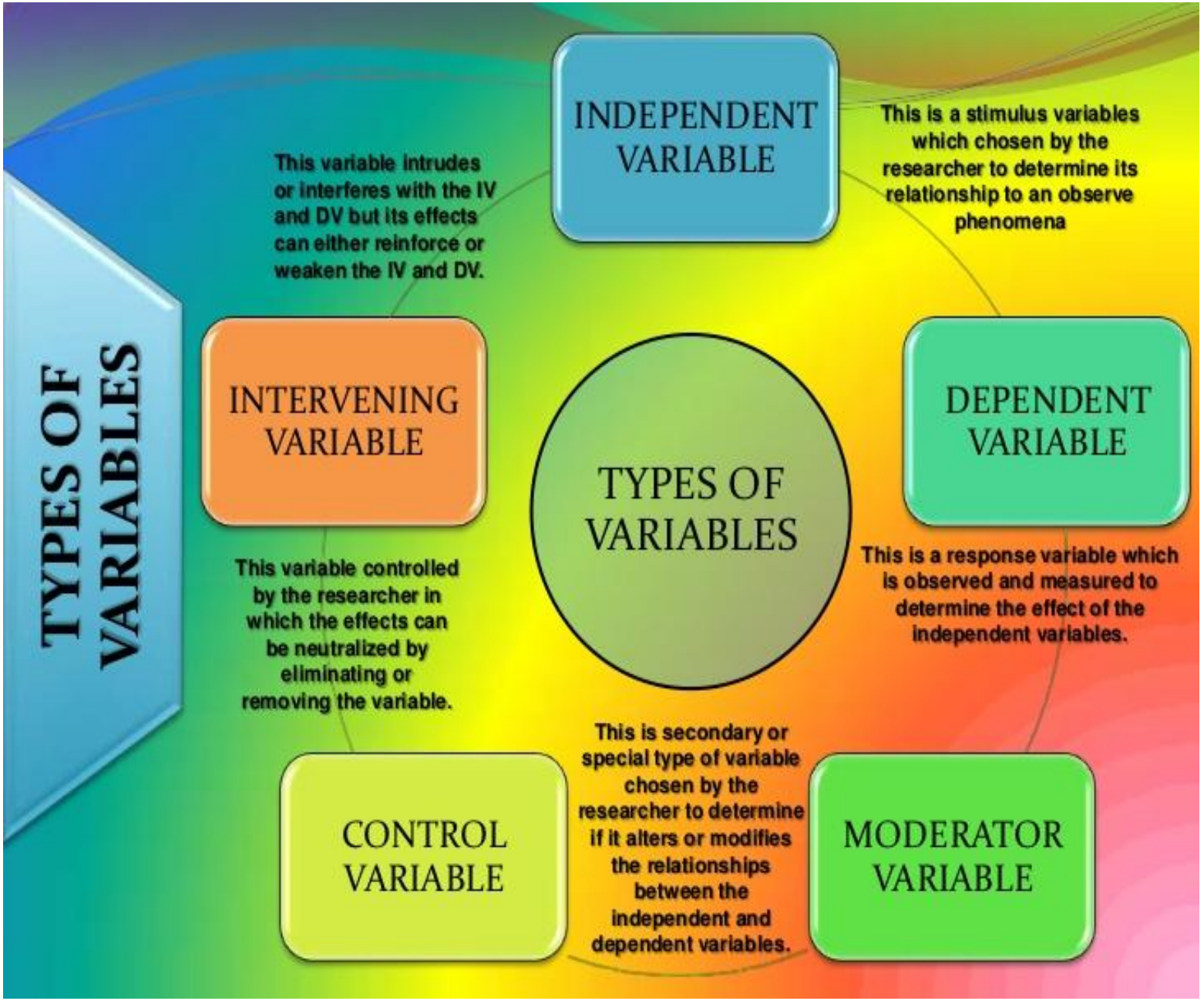

IMPORTANCE OF RESEARCH
To serve man and provide a good life
Discover more facts about unknown phenomena
Find answer to the problems by existing methods and information
Improve existing techniques and develop new instruments or products
Discover unrecognized substance
Discover pathways of action
Order related, valid generalization into systematized science
Provide basis for decision making
To satisfy researcher's curiosity
To find answers to queries
Acquire better and deeper understanding
To expand and verify existing knowledge
Improves efficiency, efficacy and quality
Improves exportations of goods
Respond to economic recovery & austerity

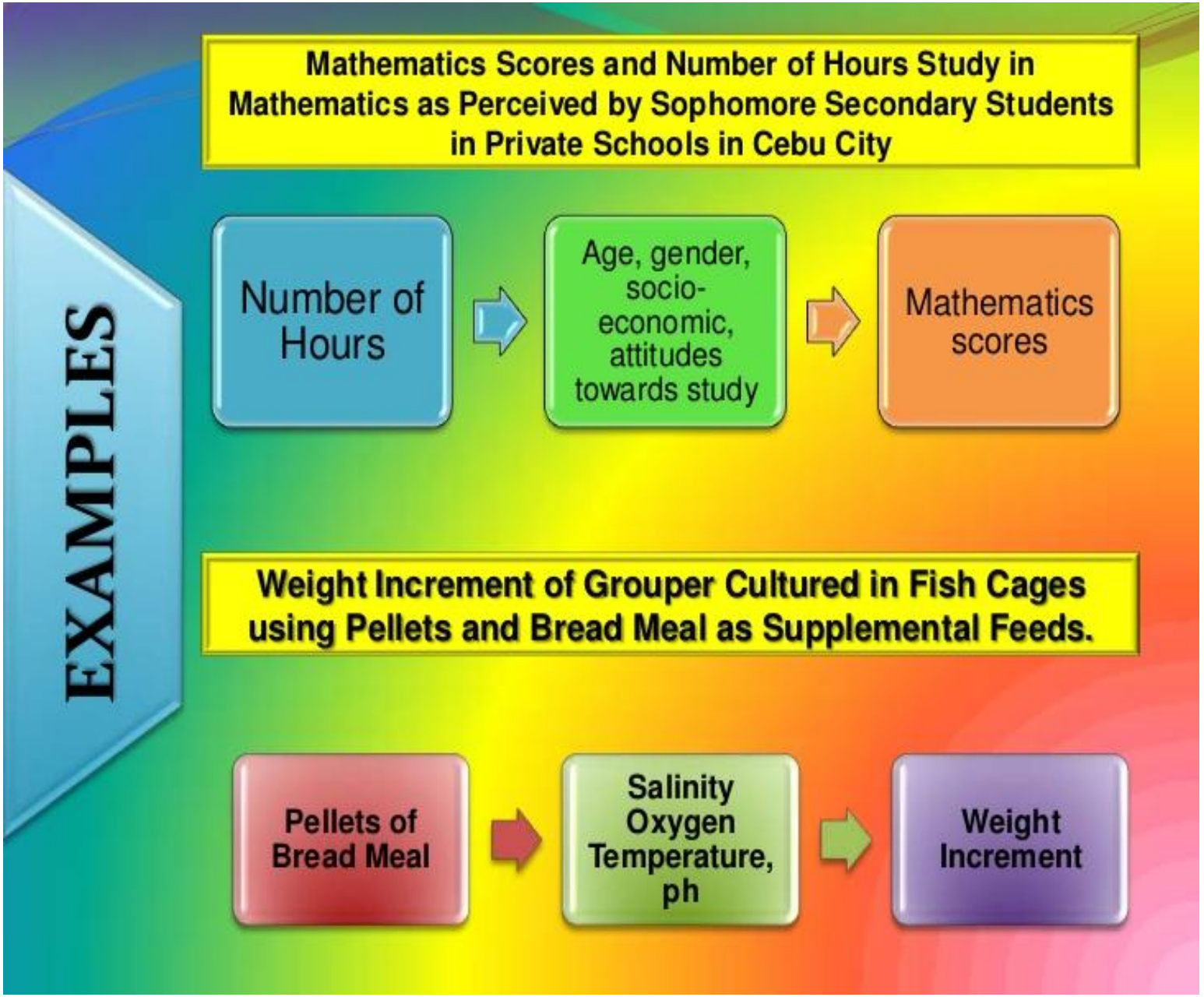

CHARACTERISTICS OF GOOD RESEARCH
EMPERICAL
Direct experience or observation
LOGICAL
Valid procedure and principles
CYCLICAL
Starts with problem & ends with problem
ANALYTICAL
Utilizes proven analytical procedures
CRITICAL
Exhibits careful and precise judgment
METHODICAL
Uses systematic method and procedures
REPLICABILITY
Repeated to arrive at valid and conclusive results

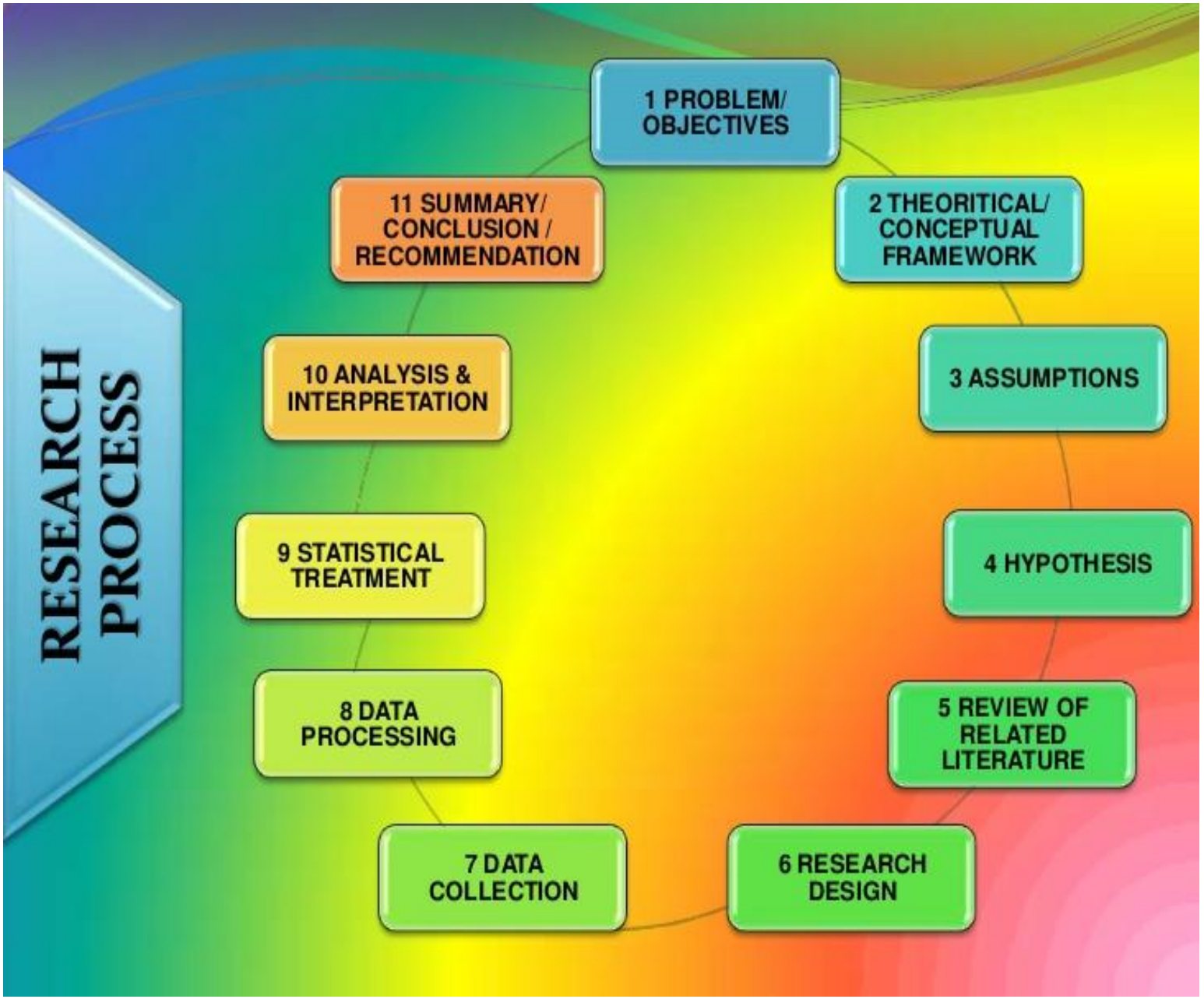

QUALITIES OF GOOD RESEARCHER
Research oriented
Efficient
Scientific
Effective
Active
Resourceful
Creative
Honest
Economical
Religious
If the research possesses this qualities, projects can be completed in short period of time and respond to socio economic recovery of the country

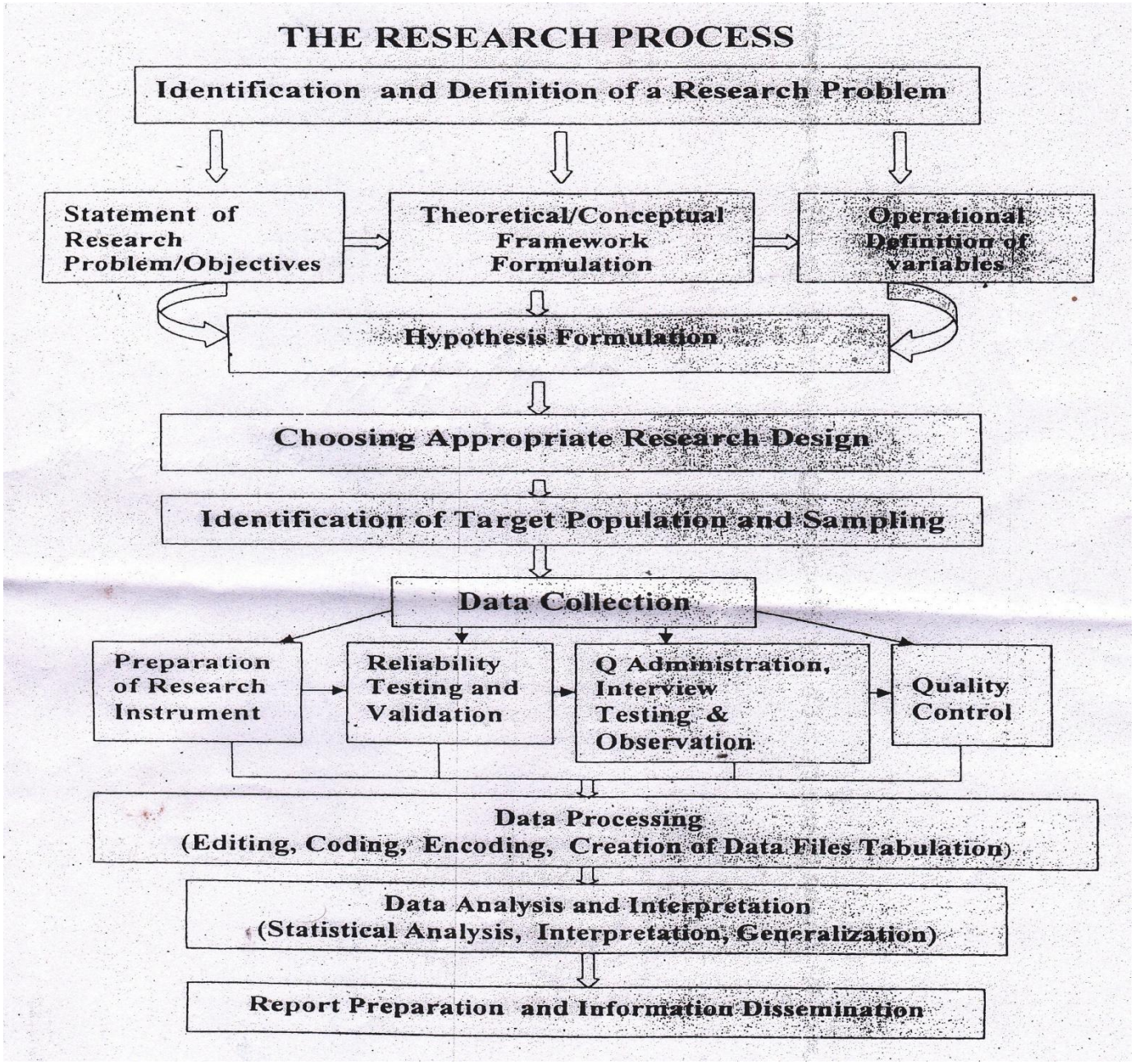

CLASSIFICATION OF RESEARCH
PURPOSE
Illuminative Research
Directive Research
Predictive Research
GOAL
Applied Research
Basic or Pure Research
LEVELS OF INVESTIGATION
Experimental
Descriptive
Exploratory
TYPE OF ANALYSIS
Holistic
Analytic
SCOPE
Action Research
CHOICE OF ANSWER
DEvelopmental
Evaluation
STATISTICAL CONTENT
Non-quantitative
Quantitative
TIME ELEMENT
Experimental
Descriptive
Historical


DEFINITION OF VARIABLES
Defined as quantity susceptible of fluctuations of change in value or magnitude under different conditions (Calmorin, 2007)
A quality or characteristics that two or more mutually exclusive values or properties (Sevilla et. All, 2002)
VARIABLE


TYPES OF VARIABLES
TYPES OF VARIABLES
INDEPENDENT VARIABLE
This is a stimulus variables which chosen by the researcher to determine its relationship to an observe phenomena
INTERVENING VARIABLE
This variable intrudes or interferes with the IV and DV but its effects can either reinforce or weaken the IV and DV.
DEPENDENT VARIABLE
This is a response variable which is observed and measured to determine the effect of the independent variables.
CONTROL VARIABLE
This variable controlled by the researcher in which the effects can be neutralized by eliminating or removing the variable.
MODERATOR VARIABLE
This is secondary or special type of variable chosen by the researcher to determine if it alters or modifies the relationships between the independent and dependent variables.


EXAMPLES
Mathematics Scores and Number of Hours Study in Mathematics as Perceived by Sophomore Secondary Students in Private Schools in Cebu City
Number of Hours
Age, gender, socio-economic, attitudes towards study
Mathematics scores
Weight Increment of Grouper Cultured in Fish Cages using Pellets and Bread Meal as Supplemental Feeds.
Pellets of Bread Meal
Salinity Oxygen Temperature, ph
Weight Increment


RESEARCH PROCESS
1 PROBLEM/ OBJECTIVES
2 THEORITICAL/ CONCEPTUAL FRAMEWORK
3 ASSUMPTIONS
4 HYPOTHESIS
5 REVIEW OF RELATED LITERATURE
6 RESEARCH DESIGN
7 DATA COLLECTION
8 DATA PROCESSING
9 STATISTICAL TREATMENT
10 ANALYSIS & INTERPRETATION
11 SUMMARY/ CONCLUSION / RECOMMENDATION


THE RESEARCH PROCESS
Identification and Definition of a Research Problem
Statement of Research Problem/Objectives
Theoretical/Conceptual Framework Formulation
Operational Definition of variables
Hypothesis Formulation
Choosing Appropriate Research Design
Identification of Target Population and Sampling
Data Collection
Preparation of Research Instrument
Reliability Testing and Validation
Q Administration, Interview Testing & Observation
Quality Control
Data Processing
(Editing, Coding, Encoding, Creation of Data Files Tabulation)
Data Analysis and Interpretation
(Statistical Analysis, Interpretation, Generalization)
Report Preparation and Information Dissemination

QUALITIES OF A GOOD RESEARCH INSTRUMENT

- Validity

  - It means the degree to which a test or measuring instrument measures what it intends to measure.

- Types of Validity

  Content validity – It means the extent to which the content or topics of the test is truly representative of the content of the course.

  Concurrent validity – It is the degree to which the test agrees or correlates with a criterion set up as an acceptable measure.

  Predictive validity – It is determined by showing how well predictions made from the test are confirmed by evidence gathered at some subsequent time.

  Construct validity – It is the extent to which the test measures a theoretical construct or trait.

- **Reliability**

  - It means the extent to which a research instrument is dependable, consistent and stable.

- **Methods in Testing the Reliability of a Good Research Instrument**

· **Test-retest method** – In this method, the same instrument is administered twice to the same group of subjects and the correlation coefficient is determined.

$$r_s = 1 - \frac{Z6D^2}{N^3-N}$$

Parallel-forms method – Parallel or equivalent forms of a test may be administered of the group of subjects, and the paired observations correlated

· **Split-half method** – The test in this method may be administered twice, but the test items are divided into two values.

$$r_{wt} = \frac{Z(r_t)}{1 + r_{nt}}$$

· **Internal-consistency method** – This method is used with psychological test which consist of dichotomously scored items.

Kuder-Richardson Formula

$$R_{xx} = \frac{[N]}{N-1}\left[\frac{SD^2 - \Sigma p_i q_i}{SD^2}\right]$$

Usability

- It means the degree to which the research instrument be satisfactorily used by teachers, researchers, supervisors and school managers without expenditures of time, money and effect.

Factors to Determine Usability

1. Base of administration. To facilitate the administration of a research instrument, instruction should be complete and precise.
2. Base of scoring. It depends on the following aspects:

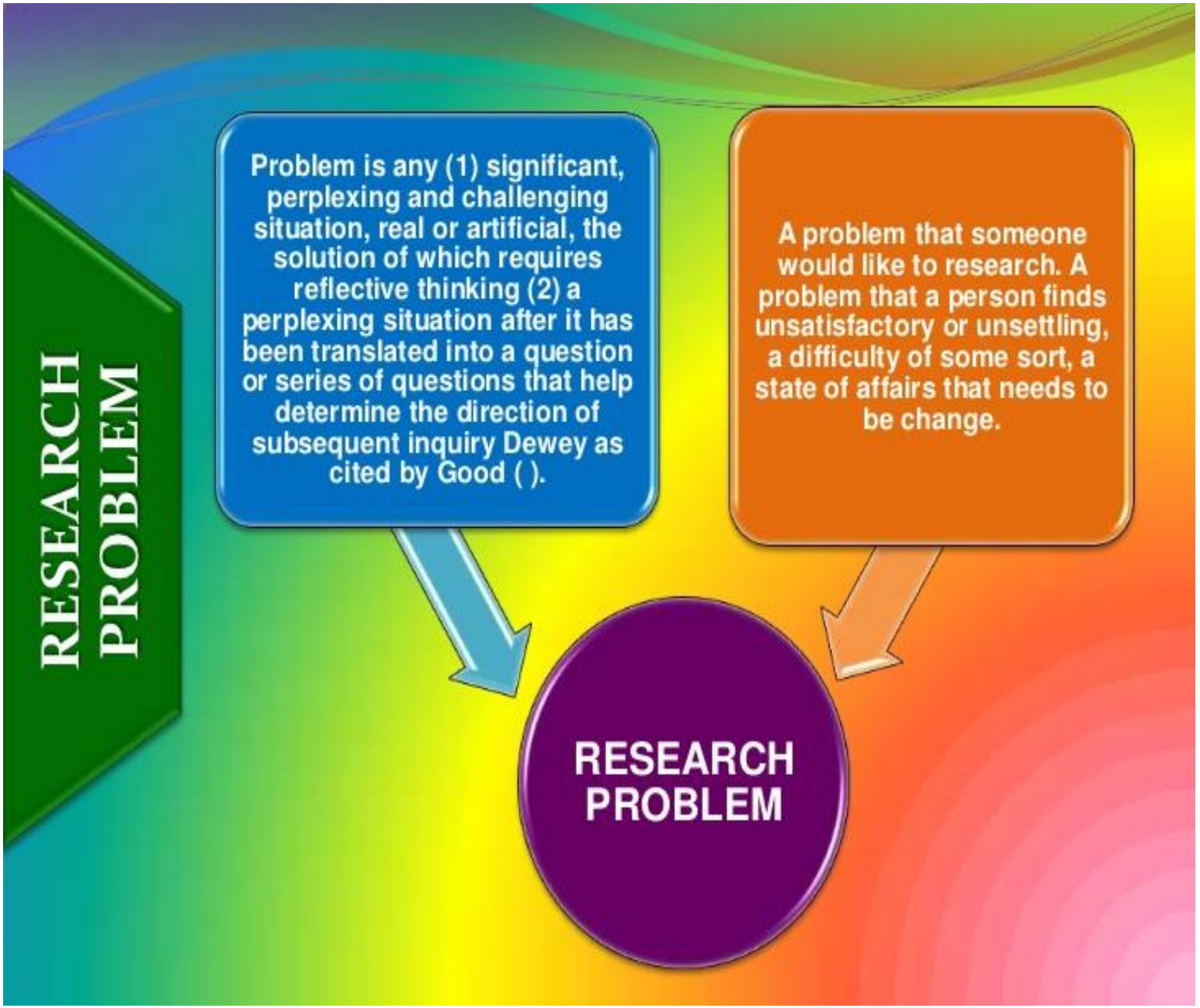
· Construction of a test in the objective type.

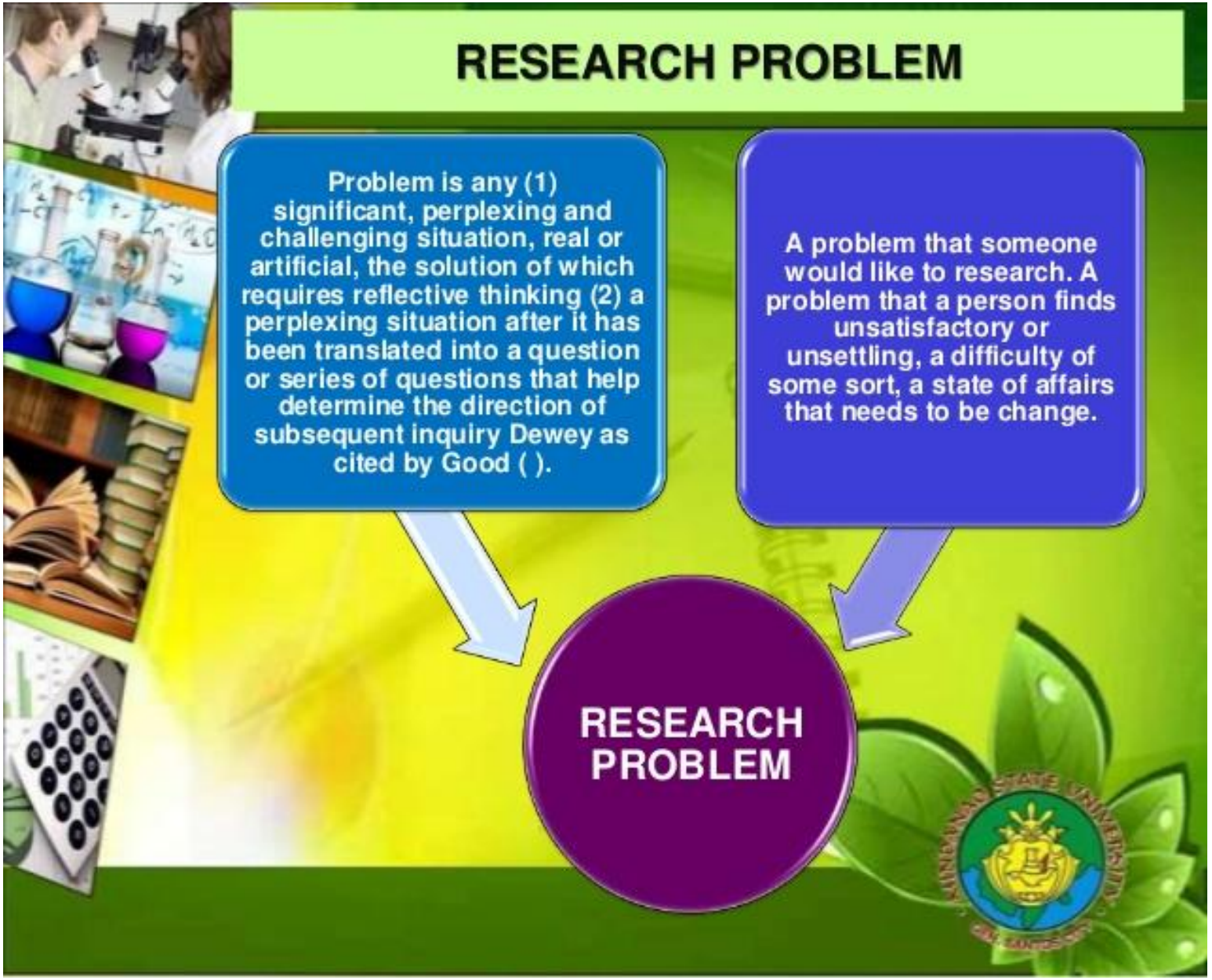
· Answer keys are adequately prepared.

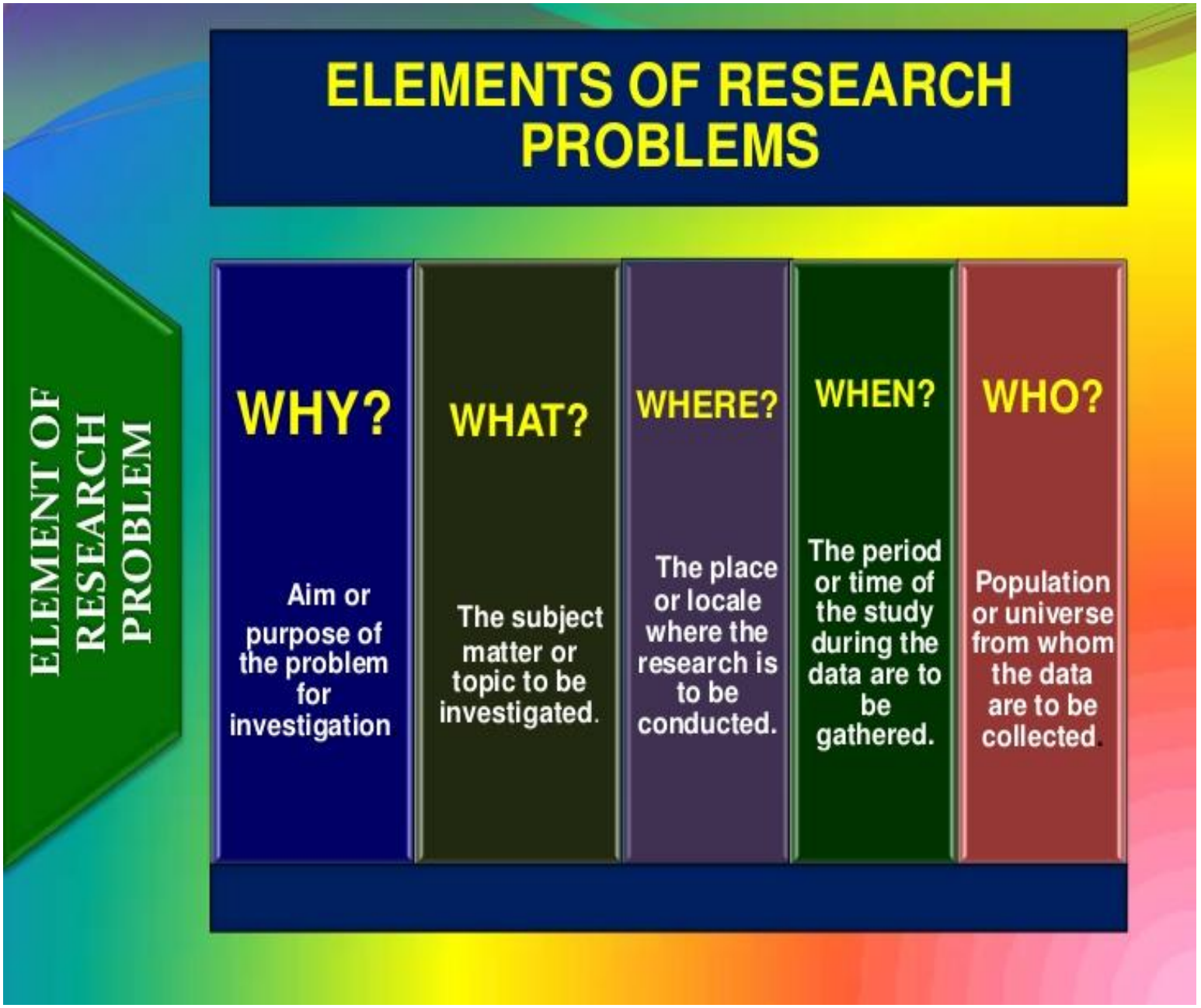
· Scoring directions are fully understood.

3. Ease of interpretation and application. Results of test are easy to interpret and apply if tables are provided.
4. Low cost. It is more practical if the test is low cost material-wise.
5. Proper mechanical make-up. A good research instrument should be printed clearly in an appropriate size for the grade or year level for which the instrument is intended.


LESSON 2
RESEARCH PROBLEM


RESEARCH PROBLEM
Problem is any (1) significant, perplexing and challenging situation, real or artificial, the solution of which requires reflective thinking (2) a perplexing situation after it has been translated into a question or series of questions that help determine the direction of subsequent inquiry Dewey as cited by Good ( ).
A problem that someone would like to research. A problem that a person finds unsatisfactory or unsettling, a difficulty of some sort, a state of affairs that needs to be change.
RESEARCH PROBLEM


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A problem that someone would like to research. A problem that a person finds unsatisfactory or unsettling, a difficulty of some sort, a state of affairs that needs to be change.
RESEARCH PROBLEM


ELEMENTS OF RESEARCH PROBLEMS
ELEMENT OF RESEARCH PROBLEM
WHY?
Aim or purpose of the problem for investigation
WHAT?
The subject matter or topic to be investigated.
WHERE?
The place or locale where the research is to be conducted.
WHEN?
The period or time of the study during the data are to be gathered.
WHO?
Population or universe from whom the data are to be collected.

Characteristics of Research Problem



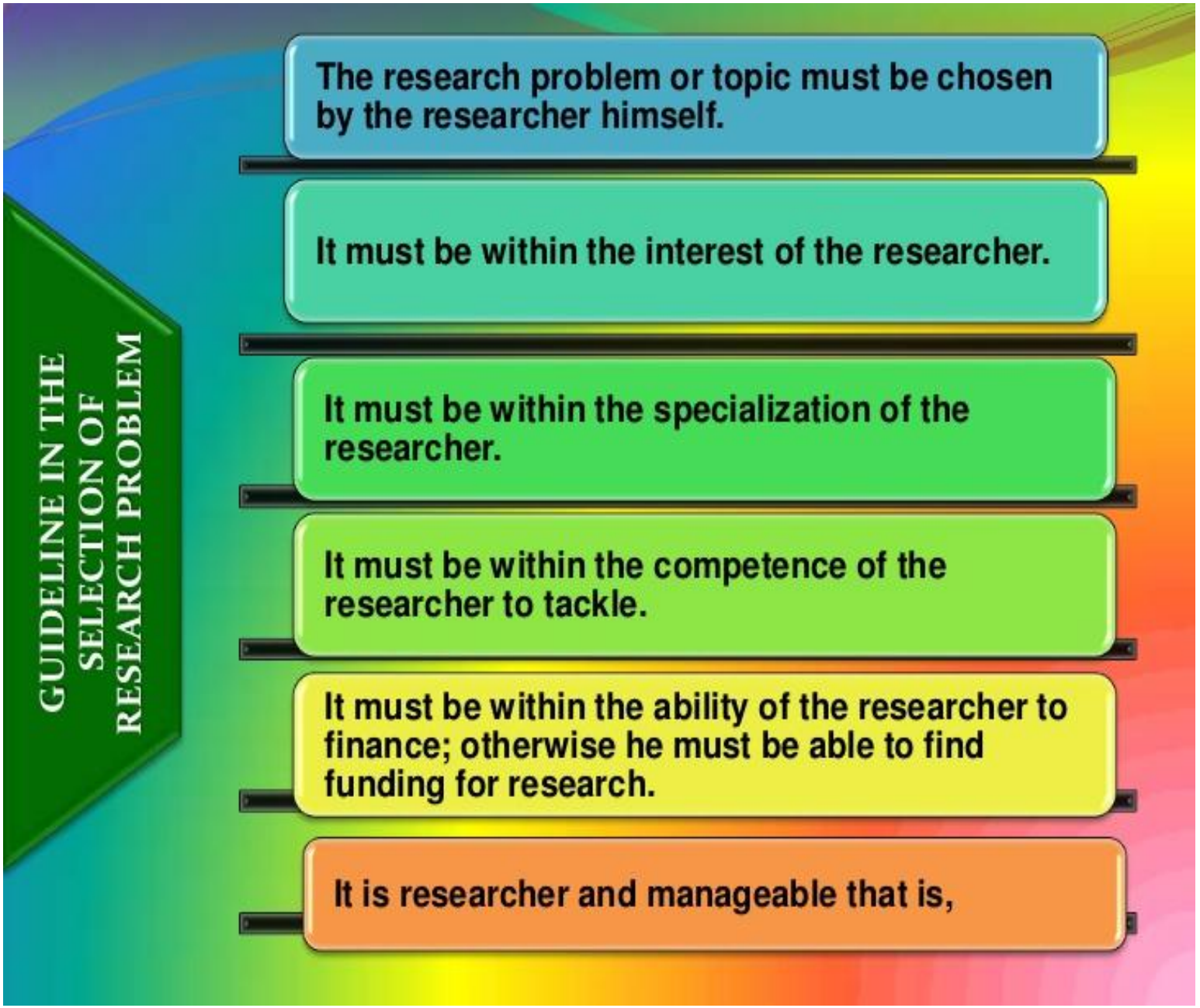
CHARACTERISTICS OF GOOD RESEARCH

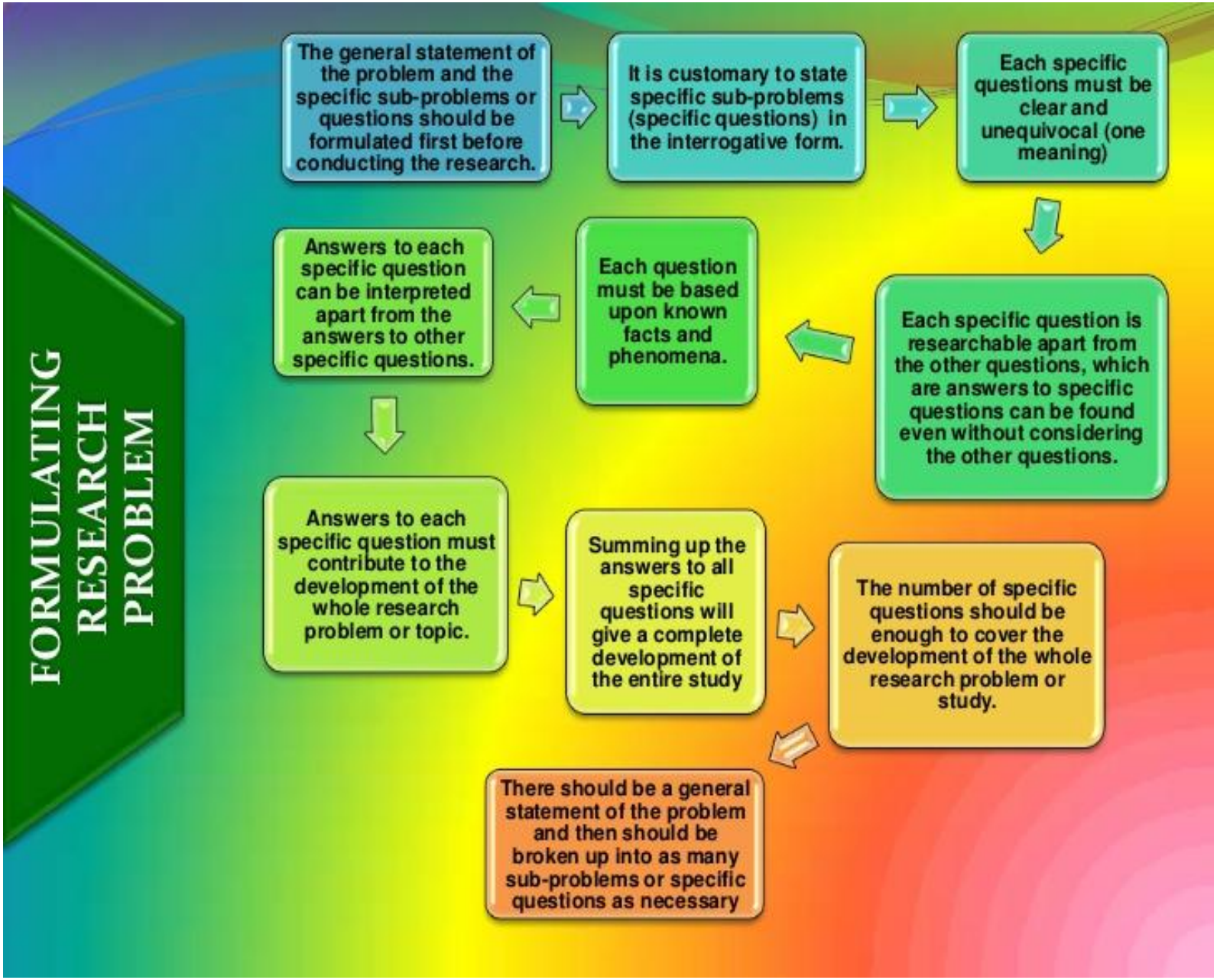
- Novelty
- Interesting
- Relevant
- Feasible
- Researchable (SMART)
- Ethical
- Systematic



Sources of Research Problems (
Calmorin & Calmorin, 2007)

1. **Specialization of the researcher**
   - The field of speicalization of the researcher is an impetus towards conduct of research
2. **Cxurrent and Past Researches**
   - Current and past are rich sources of RP
3. **Recommendations from Theses and research journals**
4. **Original and creative ideas of the researcher based on problems met in locality**


GUIDELINE IN THE SELECTION OF RESEARCH PROBLEM
The research problem or topic must be chosen by the researcher himself.
It must be within the interest of the researcher.
It must be within the specialization of the researcher.
It must be within the competence of the researcher to tackle.
It must be within the ability of the researcher to finance; otherwise he must be able to find funding for research.
It is researcher and manageable that is,


FORMULATING RESEARCH PROBLEM
The general statement of the problem and the specific sub-problems or questions should be formulated first before conducting the research.
It is customary to state specific sub-problems (specific questions) in the interrogative form.
Each specific questions must be clear and unequivocal (one meaning)
Each specific question is researchable apart from the other questions, which are answers to specific questions can be found even without considering the other questions.
Each question must be based upon known facts and phenomena.
Answers to each specific question can be interpreted apart from the answers to other specific questions.
Answers to each specific question must contribute to the development of the whole research problem or topic.
Summing up the answers to all specific questions will give a complete development of the entire study
The number of specific questions should be enough to cover the development of the whole research problem or study.
There should be a general statement of the problem and then should be broken up into as many sub-problems or specific questions as necessary



TIPS IN WRITING RESEARCH TITLE

(Bermudo et.al., 2010)

1. It must grab the attention and makes you want to read further
2. It must be intriguing, thought-provoking
3. It must include the 5Ws
4. It must be brief and concise with a maximum of 20 words and writen in inverted pyramid and in capital letters
5. It must avoid a study of, an investigation of , an analysis,,,
6. It must follow the format required by institution.



ENRICHMENT ACTIVITY

WORKSHOP 2

PROBLEM IDENTIFICATION

NAME: _____ SCORE: _____

YEAR & SECTION: _____ DATE: _____

Instructions: Give and identify your research problem by answering the given questions.

1. My three (3) probable research problems that I would like to pursue are

   a. _____
      _____

   b. _____
      _____

   c. _____
      _____

2. Among the above mentioned problems or topics, gather factors, findings or previous studies that have been reported concerning the identified problems. Collect five literature or studies taken from books, magazines, journals, thesis or dissertations, on-line resources related to problems mentioned in statement 1. Reflect the sources indicating the author's name, title of the book or research, publication, publishing company, year published

_____

_____

_____

_____

_____

_____



ENRICHMENT ACTIVITY

3. Using the collected literatures describe its nature using the matrix below.

TITLE	AUTHOR	WHY (Aim or purpose of the problem)	WHAT (the subject matter or topic to be investigated)	WHERE Place or locale where the research is to be conducted	WHEN Period or time of the study during w/s the data are to be gathered	WHO Population or universe from whom the data are to be collected
1						
2						
3						

4. My probable questions that are related to my selected problems, which remain unanswered are:

A. _____

B. _____

C. _____

5. My finalized research problem and questions which I'm most interested in:

_____

_____

_____

_____

WORKSHOP 3

ANALYZING & FRAMING RESEARCH PROBLEMS

NAME: _____ SCORE: _____

YEAR & SECTION: _____ DATE: _____

For items 1-1: Select three (3) Local Research Problem and describe its nature using the matrix below (.20 points)

TITLE	AUTHOR	WHY (Aim or purpose of the problem)	WHAT (the subject matter or topic to be investigated)	WHERE Place or locale where the research is to be conducted	WHEN Period or time of the study during w/s the data are to be gathered	WHO Population or universe from whom the data are to be collected
1						
2						
3						

For items 4-5: Describe your desired research problems using the Table below.

TITLE	WHY (Aim or purpose of the problem)	WHAT (the Variable)	WHAT (Sub-Variable)	WHERE (Place or locale where the research is to be conducted)	WHEN (Period or time of the study during w/s the data are to be gathered)	WHO (Population or universe from whom the data are to be collected)
1						
2						

STATEMENT OF RESEARCH PROBLEM/OBJECTIVES

- Research problem and research objectives have the same characteristics but they differ in form because the former is stated in interrogative or question form and the latter, in declarative form.

THE HYPOTHESES

- Hypothesis – is defined as a wise guess that is formulated and temporarily adopted to explain the observed facts covered by the study.
- Types of Hypothesis
- · Null Hypothesis – is a denial of an existence of a trait, characteristic, quality, value, correlation or difference of the result.
- · Alternative Hypothesis – is an affirmation of the existence of the observed phenomenon.

THEORETICAL FRAMEWORK

- - shapes the justification of the research problem/research objectives in order to provide the legal basis for defining its parameters

CONCEPTUAL FRAMEWORK

- - presents specific and well-defined concepts which are called constructs.

ASSUMPTIONS

- are presumed to be true statements of facts related to the research problem.

SIGNIFICANCE OF THE STUDY

- - Is presented comprehensively to convince the screening committee the importance of the study.

SCOPE AND LIMITATIONS OF THE STUDY

- - This includes the coverage of the study area, the subjects, the research apparatus, equipment or instrument, the research issues and concerns, the duration of the study, and the constraints that have direct bearing on the result of the study.

DEFINITION OF TERMS

- 1. Conceptual Definition – The definition are based on concepts or hypothetic ones which are usually taken from a dictionary, encyclopedia and published journals.
- 2. Operational Definition – The definition of terms are based on observable characteristics and how it is used in the study.

SCOPE AND LIMITATIONS OF THE STUDY

- This includes the coverage of the study area, the subjects, the research apparatus, equipment or instrument, the research issues and concerns, the duration of the study, and the constraints that have direct bearing on the result of the study.

RELATED LITERATURE AND STUDIES

Lesson 3

RELATED LITERATURE

Related literature is composed of discussions of facts and principles to which the present study is related.

The materials are usually printed and found in books, encyclopedias, professional journals, magazines, newspapers, and other publications.

IMPORTANCE, PURPOSE AND FUNCTIONS OF RELATED LITERATURE AND STUDIES

Reviewed literature and studies help or guide the researcher in the following ways:

- Help or guide the researcher in searching for or selecting a better research problem or topic.
- Help the investigator understand his topic for research better.
- Ensure that there will be no duplication of other studies.

- Help and guide the researcher in locating more sources of related information.
- Help and guide the researcher in making his research design especially in

  a. The formulation of specific questions to be researched on;

  b. The formulation of assumptions and hypotheses if there should be any;

  c. The formulation of conceptual framework;

  d. The selection and application of the methods of research;

  e. The selection and application of sampling technique;

  f. The selection and/ or preparation and validation of research instruments for gathering data;

g. The selection and application of statistical procedures;

h. The analysis, organization, presentation, and interpretation of data;

i. The making of the summary of findings, conclusions, and recommendations; and

- Help and guide the researcher in making comparison between his findings with the findings of other researchers on similar studies with the end in view of formulating generalizations or principles which are the contributions of the study to the fund of knowledge.

CHARACTERISTICS OF RELATED LITERATURE

- The surveyed materials must be as recent as possible

  Exceptions:

  A. Treatises that deal on universals or things of more or less permanent nature may still be good today.

  B. When a comparison or contrast is to be made between the conditions of today and those of a remote past, say ten or twenty years ago.

- Materials reviewed must be objective and unbiased.
- Materials surveyed must be relevant to the study.
- Surveyed materials must have been based upon genuinely original and true facts or data to make valid and reliable.
- Reviewed materials must not be too few or too many.

SOURCES AND WHERE TO LOCATE SOURCES

- Literature reviewed typically includes scholarly journals, scholarly books, authoritative databases and primary sources. Sometimes it includes newspapers, magazines, other books, films, and audio and video tapes, and other secondary sources.

- **Primary sources** are the origin of information under study, fundamental documents relating to a particular subject or idea. Often they are first hand accounts written by a witness or researcher at the time of an event or discovery.

E.g. published books, magazines, encyclopedias, almanacs and etc.

- Secondary sources are documents or recordings that relate to or discuss information originally presented elsewhere.

E.g. Information from the internet, unpublished thesis and dissertations

WHERE TO LOCATE THE SOURCES?

Generally, the sources of related literature and studies are located in the following places:

- Libraries, either government, school, or private libraries.
- Government and private offices.
- The National Library.
- The library of the Department of Education, Culture and Sports.

The last two are especially rich depositories of related materials, particularly unpublished master's thesis and doctoral dissertations.

RELATED LEGAL BASES

Related – means the legal bases, literature and studies which have direct bearing or relation to the present study.

It determine the relevance of the study to the government's thrusts. The major sources of related legal bases are laws and department directives such as circulars, orders, memoranda, and others which are related to the present study.

- **Example:**
- *Republic Act No. 7164 known as the "Philippine Nursing Act of 1991" defines professional nursing as the performance for a fee, salary, or other reward or compensation… toward the promotion of health, and alleviation of suffering through utilization of nursing process.* ***(LEGAL BASIS)*****.**
- *The foregoing professional nursing definition explicitly expressed that nurses practice their profession for a fee or salary commensurate with their work in order to satisfy their needs and wants to cope with the fast changing world. In other words, the performance of a particular job affects nurses' values especially if nurses meet the problems related to their job in the hospital, hence, promotion of health services cannot be attained.* ***(EXPLANATION OF THE LEGAL BASIS)***

RELATED LITERATURE

- It is a section in a research paper, thesis dissertation, and research project in which the sources are taken from books, journals, magazines, novels, poetry, etc that contains facts, laws, theories and other documented observations. This is in a chronological order from recent to past when presented. It is unscientific if related literature are presented and of no explanation at the relevance to the present study.

- **Example:**
- Murray's (1998) article on nurse executives' leadership roles stated that chief nursing officers (CNO) leave their position due to lack of power conflict with chief executive officer (CEO), and inadequate nursing personnel. **[Related literature by Bruce P. Murray. 1998. "Nurse Executives' Leadership Roles." Journal of Nursing Administration. 28(6):48. (June 1998)].**
- The article of Murray has bearing to the present study because staff nurses' performance would be affected if the middle managers or chief nursing officers have conflict with the chief executive officer or top manager and with problems met related to their job, for instance, inadequate nursing personnel. **(EXPLANATION OF THE RELATED LITERATURE)**

RELATED STUDIES (LOCAL OR FOREIGN)

- Published and unpublished research studies are sources of materials that are included in this section. The research studies which have direct bearing to the present study are segregated into foreign and local studies.

- **Example of LOCAL STUDY:**
- In 1999, Sultan in her study on perceived sources of stress among staff nurses at a tertiary hospital in Quezon City, found out that lack of administrative support and no rewards were perceived by staff nurses as the most stressing… **(Related local study by Vilma Sultan. 1999. "Perceived Sources of Stress Among Staff Nurses at Tertiary Hospital in Quezon City." MA Thesis in Nursing. Central Philippine University, Iloilo City, Philippines.)**
- Sultan's study has bearing to the present study because lack of administrative support, no rewards, and no opportunities for career development are among the job-related problems met by staff nurses in private and government hospitals in Iloilo City which block their way to perform effectively and efficiently. **(EXPLANATION OF THE RELATED STUDY)**

CONCEPTUAL AND OPERATIONAL

Conceptual - also known as constitutive, is that which is given in dictionaries. It is the academic or universal meaning attributed to a word or group of works. Moreover, it is mostly abstract and formal in nature.

Operational – also known as functional. Operational definition may be measured and experimental. The measured operational definition states the way the concept is measured in the investigation. In an experimental operational definition the researcher may spell out the details of the manipulation of a variable.

RESEARCH METHODS/RESEARCH DESIGNS

Lesson 4

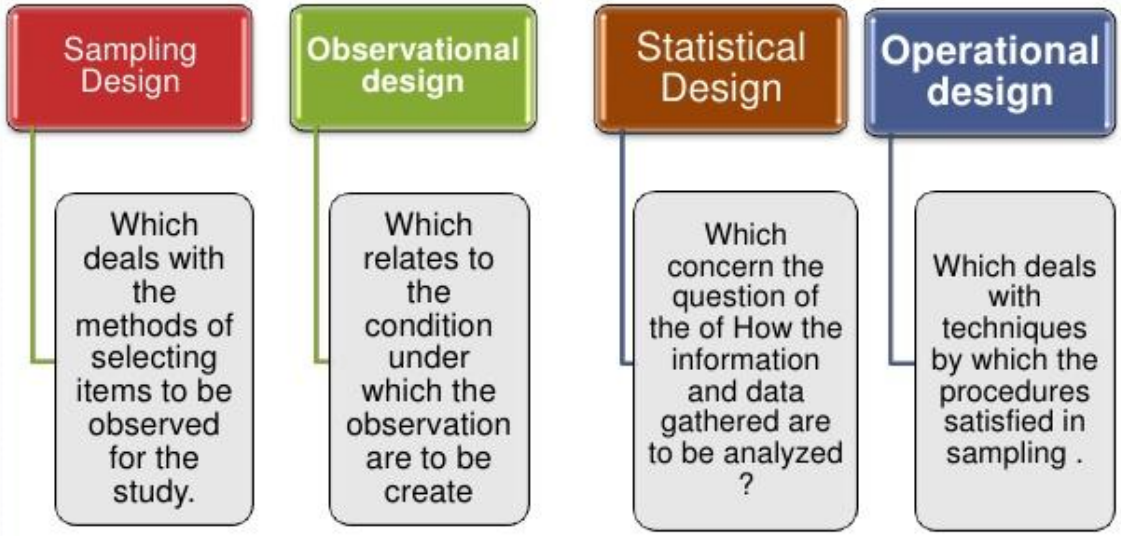
WHAT IS RESEARCH DESIGN ?

- Task of defining the research problem is the preparation of the research project, popularly known as the “research design".
- Decisions regarding **what, where, when, how much, by what means concerning an inquiry or a research study constitute a research design.**

Meaning of research design

1	2	3
• A research design is the ***arrangement of conditions for collection and analysis of data*** in a manner that ***aims to combine relevance to the research purpose with economy in procedure.***	• Blueprint of the study. It guides the collection, measurement and analysis of data	• A plan or structure of the investigation in order to obtain answers to research questions. It outline what the researcher would like to do until the final analysis of the data

Parts of Research Design



Features of Good RS

A research design appropriate for a particular research problem, usually involves the following features.

The mean of obtaining information.

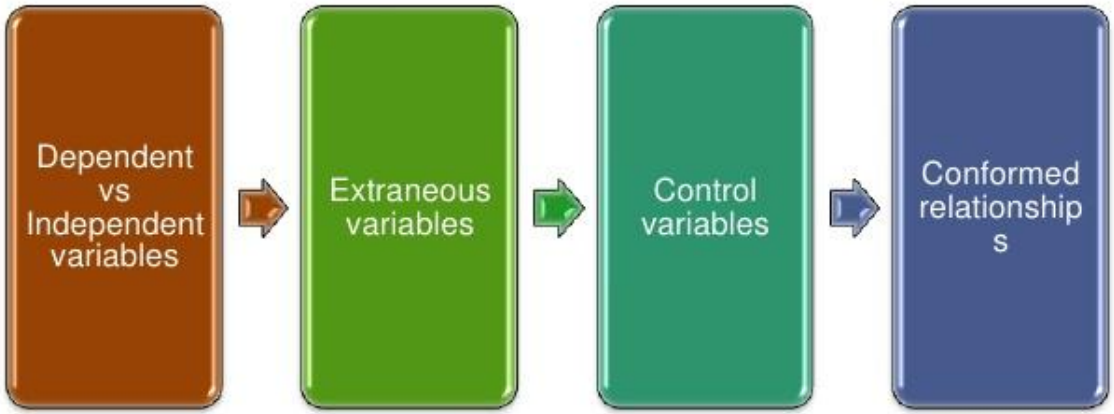
The availability and skills of the researcher and his staff, if any.

The objective of the problem to be studied.

The nature of the problem to be studied .

The availability of time and money for the research work.

Important Concepts of RS



Research hypothesis

- The researcher hypothesis is a predicative statement that relates an independent variable to dependent variable.

Experimental and non-experimental hypothesis testing

When the purpose of research is to test a research hypothesis, it us termed as hypothesis testing research .It can be experimental or non-experimental

Experimental and control groups

- When a group is exposed to usual conditions, it is termed as a control group.
- But when the group is exposed to be some special condition, it is termed as Experimental group

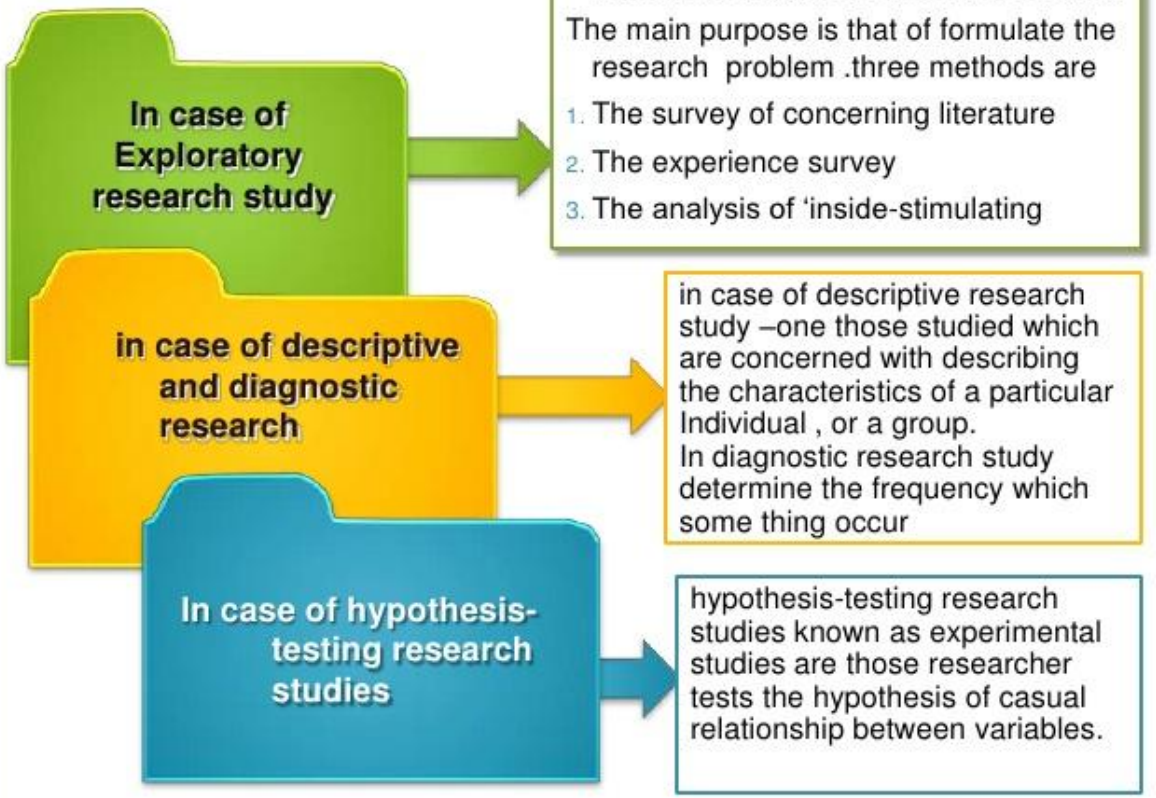
Treatments

- The different conditions under which Experiment and control groups are put up usually referred to as treatment.

Experiment

- The process of examining the truth of a statistical hypothesis, relating to some research problem, is known as an Experiment.

Different research design



Difference between formulative= descriptive/diagnostic

- Flexible design
- Judgmental sampling
- No pre-determined design
- No fixed decision about the operational procedures

- No flexibility
- Random sampling
- Pre-determined design for analysis
- Advanced decisions

Research Designs/Approaches

Type	Purpose	Time frame	Degree of control	Examples
Experimental	Test for cause/ effect relationships	current	High	Comparing two types of treatments for anxiety.
Quasi-experi-mental	Test for cause/ effect relationships without full control	Current or past	Moderate to high	Gender differences in visual/spatial abilities
Non-experimental - correlational	Examine relationship between two variables	Current (cross-sectional) or past	Low to medium	Relationship between studying style and grade point average.
Ex post facto	Examine the effect of past event on current functioning.	Past & current	Low to medium	Relationship between history of child abuse & depression.

Research Designs/Approaches

Type	Purpose	Time frame	Degree of control	Examples
Non-experimental correlational	Examine relationship bet 2 var. where 1 is measured later.	Future - predictive	Low to moderate	Relationship bet history of depression & development of cancer.
Cohort-sequential	Examine change in a var. over time in overlapping groups	Future	Low to moderate	How mother-child negativity changed over adolescence
Survey	Assess opinions or characteristics that exist at a given time.	Current	None or low	Voting preferences before an election.
Qualitative	Discover potential relationships; descriptive.	Past or current	None or Low	People's experiences of quitting smoking.

Research Methods: Historical Research

What is Historical Research?

Historical Research is the systematic collection and evaluation of data to describe, explain, and thereby understand actions or events that occurred sometime in the past. (*J.M. Fraenkel & N.E. Wallen)*

The unique characteristic of historical research is that it focuses exclusively on past occurrences and events.
(J.M. Fraenkel & N.E. Wallen)

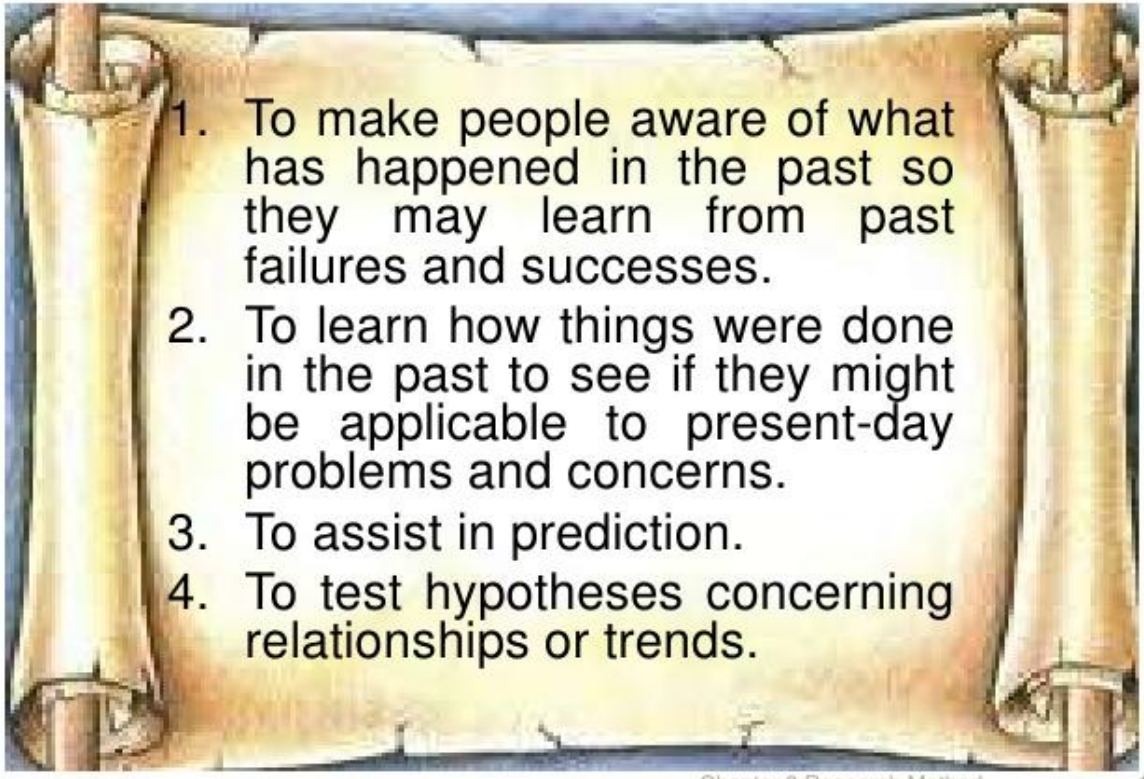
Advantages

- It is the only research method that can study evidence from the past.
- Wider range of evidence than most other methods.

Disadvantages

- The measures used in other methods to control for threats to internal validity are simply not possible in a historical study.
- Bias

USES OF HISTORICAL RESEARCH



1. To make people aware of what has happened in the past so they may learn from past failures and successes.
2. To learn how things were done in the past to see if they might be applicable to present-day problems and concerns.
3. To assist in prediction.
4. To test hypotheses concerning relationships or trends.

MAJOR STEPS OF HISTORICAL DESIGN

I. **DEFINING THE PROBLEM**
II. **LOCATING RELEVANT SOURCES**

Categories of Sources:

1. *Documents*– are written or printed materials that have been produced in some form or another—annual reports, artwork, bills, books, cartoons, circulars, court records, diaries, diplomas, legal records, newspapers, magazines, notebooks, school yearbooks, memos, tests , and so on.

2. *Numerical records*. Numerical, or *quantitative,* records can be considered either as a separate type of source in and of themselves or as a subcategory of documents—test scores, attendance figures, census reports, school budgets, and the like.

MAJOR STEPS OF HISTORICAL DESIGN

3. ***Oral statements***– another valuable source of information for the historical researcher lies in the statements people make orally—stories, myths, tales, legends, chants, songs and other forms of oral; expression have been used by people down through the ages to leave a record for future generations.

4. ***Relics-*** is any object whose physical or visual characteristics can provide some information about the past—furniture, artwork, clothing, buildings, monuments, or equipment.

PRIMARY VERSUS SECONDARY SOURCES

Primary Source

Is one prepared by an individual who was a participant in or a direct witness to the event being described.

Secondary Source

Is a documents prepared by an **individual** who was not a direct witness to an event, but obtained his or her description of the event from someone else.

MAJOR STEPS OF HISTORICAL DESIGN

PRIMARY SOURCE

1. A nineteenth-century teacher's account of what it was like to live with a frontier family.
2. A photograph of an eight-grade graduating class in 1930.
3. Essays written by students in response to the question, "What do you like most and least about school?"
4. Minutes of a school board meeting taken by the secretary of the board.
5. Songs composed by members of a high school glee club.

SECONDARY SOURCE

1. An encyclopedia describing various types of educational research conducted over a 10-year period.
2. A magazine article summarizing Aristotle's views on education.
3. A student's report to her counselor of why her teacher said she was being suspended from school.
4. A book describing schooling in the New England colonies during the 1700s.
5. A newspaper account of a school board meeting based on oral interviews with members of the board.

MAJOR STEPS OF HISTORICAL DESIGN

III. SUMMARIZING INFORMATION OBTAINED FROM HISTORICAL SOURCES

IV. THE EVALUATION OF HISTORICAL SOURCES

External Criticism. Refers to the genuineness of any and all documents the researcher uses.

Several questions come to mind in evaluating the genuineness of a historical source:

- *Who* wrote this documents? Was the author living at that time?
- *For what purpose* was the document written? For whom was it intended? And why?
- *When* was the document written? Is the date on the document accurate? Could the details describes have actually happened during this time?
- *Where* was the document written? Could the details described have occurred in this location?

MAJOR STEPS OF HISTORICAL DESIGN

- *Under what conditions* was the document written? Is there any possibility that what was written might have been directly or subtly coerced?
- Do *different forms or versions* of the document exist?

Internal Criticism. Once researchers have satisfied themselves that a source document is genuine, they need to determine if the *contents* of the documents are *accurate*. This involves what is known as **internal criticism.** Both the accuracy of the information contained in a document and the truthfulness of the author need to be evaluated.

With regard to the author of the document:

- Was the author *present* at the event he or she is describing? In other words, is the document a primary or a secondary source?
- Was the author a *participant* in *or* an *observer of the event?*
- Was the author *competent* to describe the event?



Research Methods: Descriptive Design



What Is Descriptive Research?

The descriptive research is the most popular approach used in research project. Derived from the Latin word **de** meaning from and **scribe** meaning write.

Descriptive research includes all studies that purport to present facts concerning the nature and status of anything – a group of persons, a number of objects, a set of conditions, a class of events, a system of thought or any other kind of phenomena which one may wish to study (Manuel and Medici).

Descriptive research is fact-finding with adequate interpretation. It is something more and beyond just data-gathering; the latter is not reflective thinking nor research. The true meaning of the data collected should be reported from the point of view of the objectives and basic assumption of the project underway.

As defined by Best (1963) – Descriptive research describes and interprets what is. It is concerned with conditions of relationships that exist, practices that prevail, beliefs and processes that are going on, effects that are being felt, or trends that are developing.

What Is Descriptive Research?

Descriptive research can be either *quantitative* or *qualitative*. It can involve collections of quantitative information that can be tabulated along a continuum in numerical form, such as scores on a test or the number of times a person chooses to use a-certain feature of a multimedia program, or it can describe categories of information such as gender or patterns of interaction when using technology in a group situation. Descriptive research involves gathering data that describe events and then organizes, tabulates, depicts, and describes the data collection (Glass & Hopkins, 1984). It often uses visual aids such as graphs and charts to aid the reader in understanding the data distribution. Because the human mind cannot extract the full import of a large mass of raw data, descriptive statistics are very important in reducing the data to manageable form. When in-depth, narrative descriptions of small numbers of cases are involved, the research uses description as a tool to organize data into patterns that emerge during analysis. Those patterns aid the mind in comprehending a qualitative

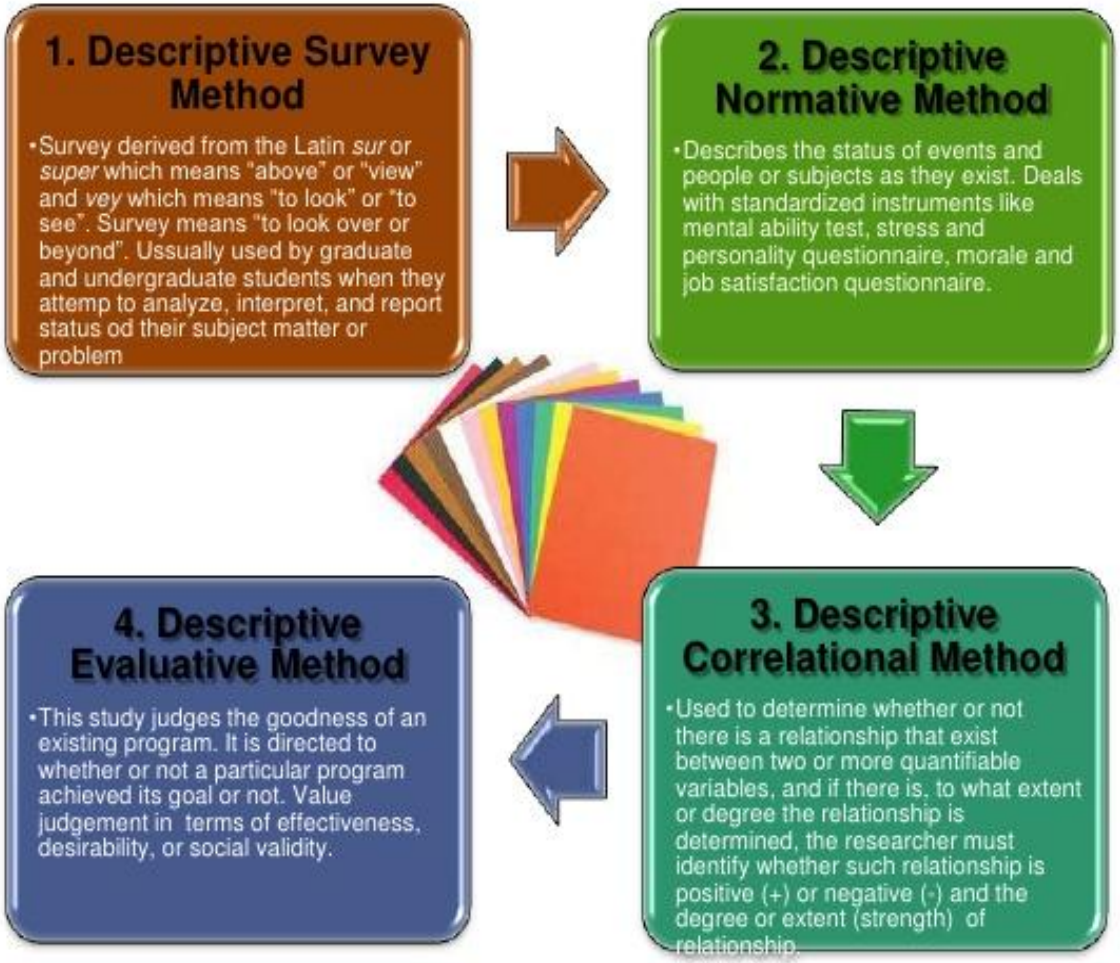
*Most quantitative research falls into two areas: studies that describe events and studies aimed at discovering inferences or causal relationships. Descriptive studies are aimed at finding out "**what is**," so observational and survey methods are frequently used to collect descriptive data (Borg & Gall, 1989). Studies of this type might describe the current state of multimedia usage in schools or patterns of activity resulting from group work at the computer. An example of this is Cochenour, Hakes, and Neal's (1994) study of trends in compressed video applications with education and the private sector.*

VALUES OF DESCRIPTIVE STUDIES

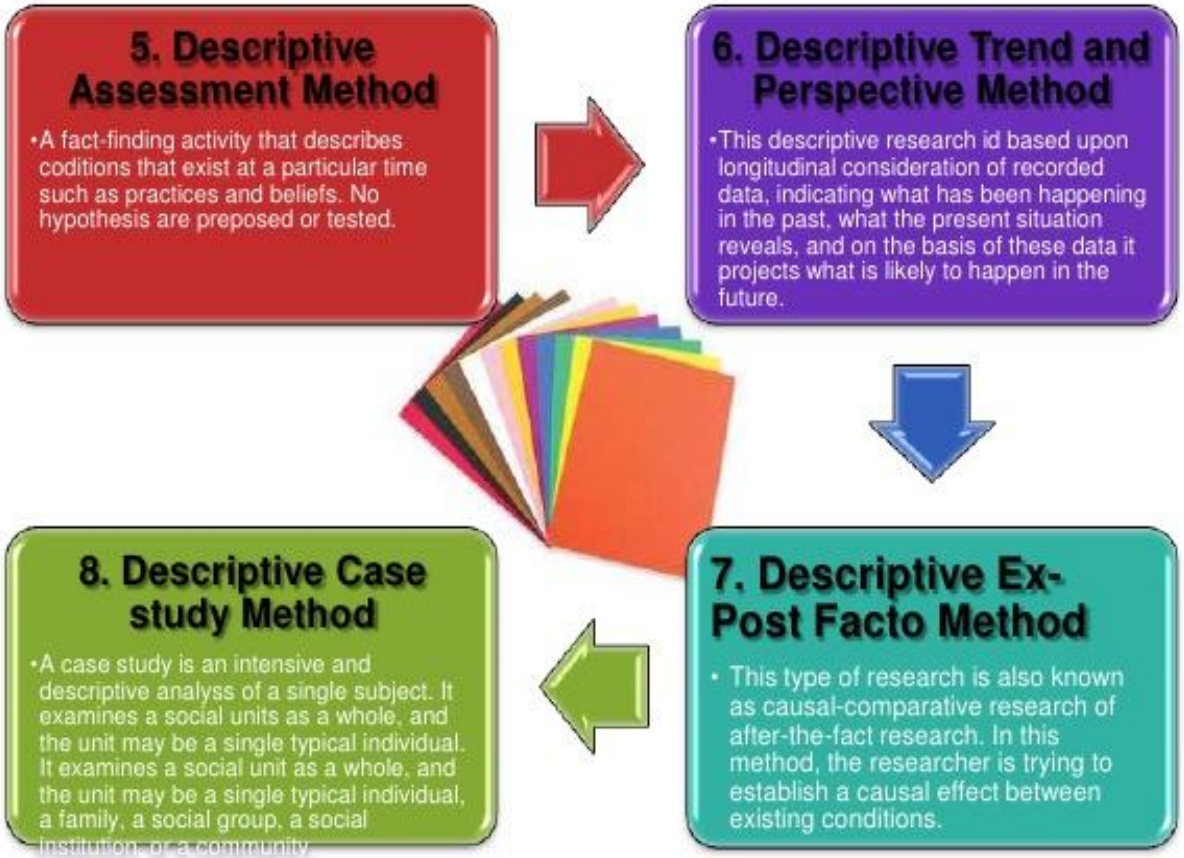
- Descriptive studies are of great value in providing facts on which professional judgements may be based
- Descriptive studies contribute to science because they afford penetrating insights into the nature of what are dealt with and what to study
- Descriptive studies contribute to science as they help fashion many o
- It contributes to the establisment of standard norms of conduct, behavior or performance.f the tools for and of research.
- It reveals problems or abnormal conditions so that remedial measures may be instituted. It reveals to us what we do not want, what we want and how to acquire what we want.
- It makes possible the prediction of the future on the basis of findings on prevailing conditions, correlationsand on the basis of reaction of the people towards certain issue.
- Provides basis for decision-making.



Types of Descriptive Research



Types of Descriptive Research



Types of Descriptive Research

9. Descriptive Comparative Method

- The researcher is knowledgeable about the operation of natural or social forceswhich maybe obtained by watching them analytically at stated intervals in their development over a long periood of time as seen desirable and possible. Also called a *genetic method.*

Descriptive Comparative Method. This method provides an explanation about the extent of relationship between two or more variables. The researcher examines the relationsships including similarities or differences among several variables.

Advantages of Comparative Study:

- Intervening variables of comparative study can easily be noted and discerned
- Research variables can easily be selected.
- Comparative results of this study can be known immediately
- Comparative method is easily conducted.

Disadvantages of Comparative Study:

- The findings of the study are very much limited because it cannot show the cause of a change in another variable.
- Variables to be selected are likewise limited.
- It is difficult to get samples which have the same characteristics or attributes.



Research Methods: Experimental Design



What is Experimental Design?

1. problem solving approach that the study is described in the future on what will be when variables are carefully controlled or manipulated
2. can be a basic, applied, or developmental research
3. a research wherein a researcher manipulates and controls one or more independent variables for variation concomitant to the manipulation of the dependent variable
4. used to find out what caused the change in the characteristics or behavior of the subjects and what change or effect has been made then



Distinguishing Features

The only design wherein the invest – harvest principle can be attained.

The control and manipulation of variables constitute the essential feature of experimental design

It is usually involves two groups – an experimental group and a control group

It is expected to reveal cause and effect relationship even if the experiment conducted in the field or laboratory



Experimental group – exposed to the influence of a factor - an intervention or treatment under consideration

Control group – not exposed to the same factor to which experimental group is exposed to, or it receives a different treatment, or it is left to its usual way of doing things

It is given priority by research institutions in giving of research grants-in-aid

Experimental design involves the **7 Ms**

- Manpower
- Money
- Methods
- Materials
- Machinery
- Moment of time
- Marketing



Experimental Plan

1. Determining the experimental problem or topic
2. Surveying related literature and studies
3. Formulating the hypothesis or hypotheses
4. Identifying the experimental variables
5. Select the appropriate experimental design
6. Selecting the sample
7. Selecting or constructing appropriate instruments to measure the results of the experiment
8. Conducting the experiment
   - ❑ History
   - ❑ Maturation
   - ❑ Testing
   - ❑ Biases in the selection of the respondents
   - ❑ Experimental mortality
   - ❑ Interaction effects
9. Measuring the results and applying statistical procedures
10. Writing the reports of the experiment



Experimental designs



1 Posttest only design

- Also known as one shot case study
- involves just one group of subjects which is exposed to an intervention or treatment. After sometime it is given a posttest or is subject for observation.

Experimental Group | X | O

2 The one-group pretest-posttest design

- As to posttest-only, this design is without a control group.
- However, a pretest or baseline observation (O1) which allows the investigator to determine the effects of the treatment by comparing pretest and posttest (O2) results.

Experimental Group | O_1 | X | O_2

Experimental designs



- It has a control or comparison group.
- Experimental group is exposed to an intervention or treatment (X) which is derived of the control group.
- After some time the experimental group and the control group are observed

3 The static-group comparison design

Experimental Group	X	O
Control Group		O

- this design involves the random assignment of subjects from a single population the experimental and the control groups.
- this two groups receive initial observations or pretests (O1 & O3) and posttest (O2 & O4) to determine the differences.

4 The pretest-posttest control group design

R Experimental Group	O_1	X	O_2
R Control Group	O_3		O_4

Experimental designs



5 The posttest-only control group design

- Involves both the experimental and the control groups which were created through random assignment (R) subjects from a study population.
- both are not given pretest; both receives posttest after the experimental group has been exposed to intervention or

R Experimental Group

R Control Group

X	O_1
	O_2

6 The Solomon four-group design

- The subjects are randomly assigned to four groups.
- Two groups are experimental and the other two groups are control groups.
- One experimental group and one control group is subject to pretest, and all four are subject to posttest.

R Experimental Group
R Control Group
R Experimental Group
R Control Group

O_1	X	O_2
O_3		O_4
	X	O_5
		O_6

Experimental designs



- involves two groups which are the experimental and the control groups.
- like pretest-posttest control group design both experimental and control group undergo pretest and posttest, the only difference is that, in this design the assignment of subjects into these groups is not done

7 The non-equivalent control group design

Experimental Group	O_1	X	O_2
Control Group	O_3	X	O_4

- involves two groups, the creation of which was done by a random selection (R) of samples.
- one group is subject to a baseline pretest (O1), and another group is used for the posttest measurement (O2).

8 Separate Sample pretest-posttest group design

R Pretest Group	O_1	X
R Posttest Group	O_2	X

Experimental designs



- an improvement of the one-group pretest-posttest design because it has the advantage of repeated observations before and after the intervention or treatment has been introduced to the study group.
- the more number of pretest and posttest the better it will be for study.

9 The time-series design

Experimental Group: $O_1\ O_2\ O_3\ O_4\ X\ O_5\ O_6\ O_7\ O_8$